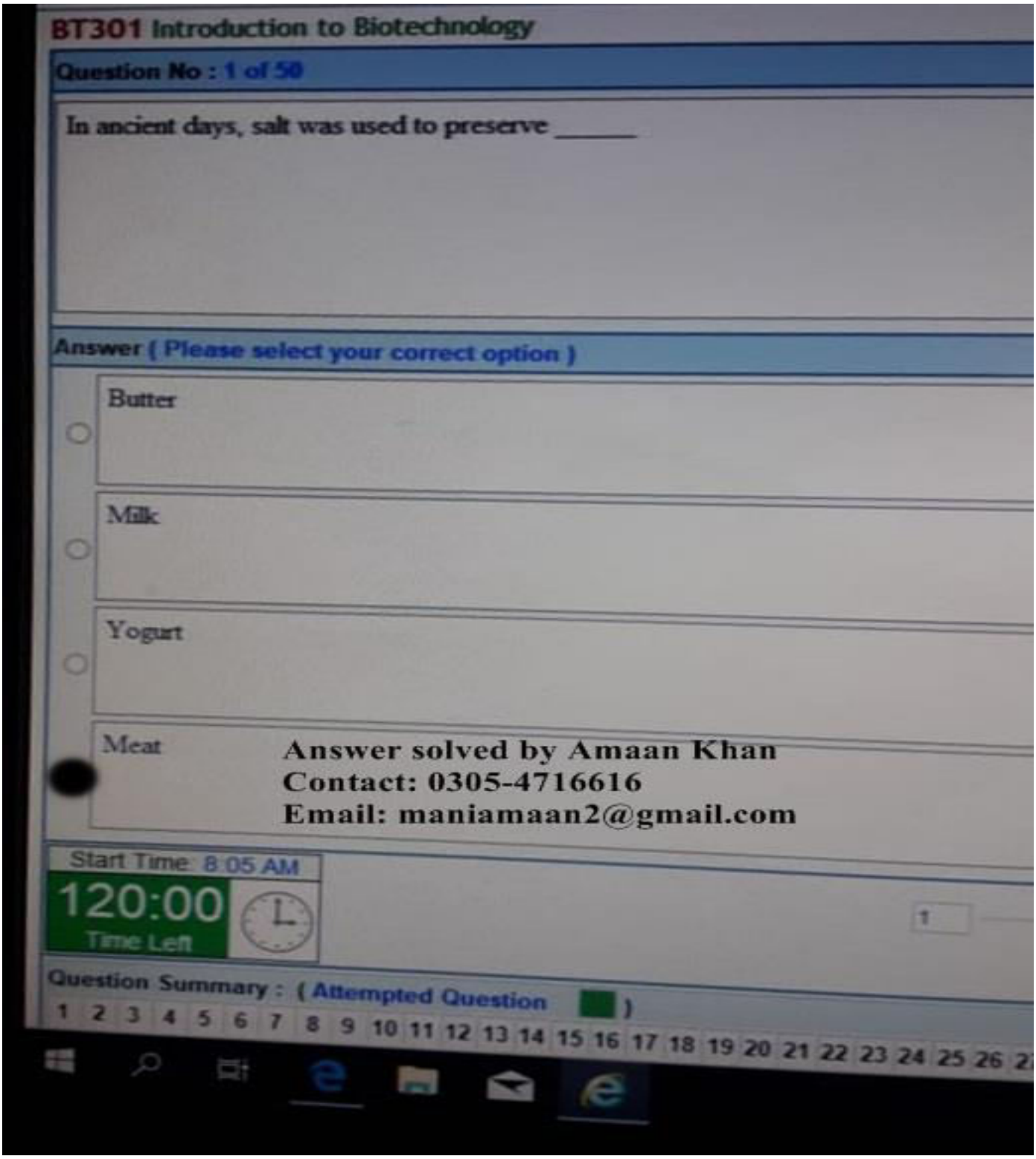
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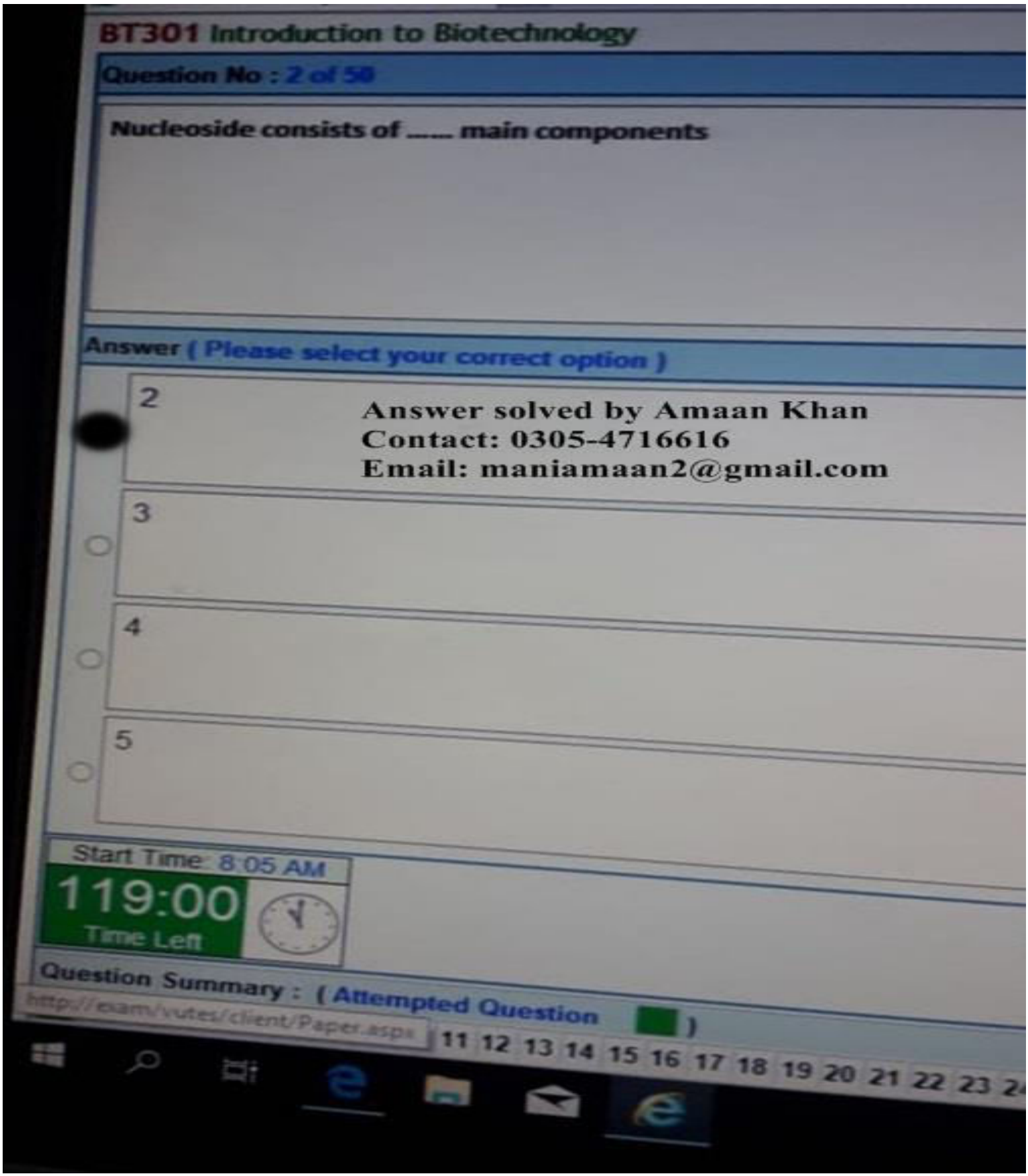
BT 301 FINAL TERMS

SPRING 2020



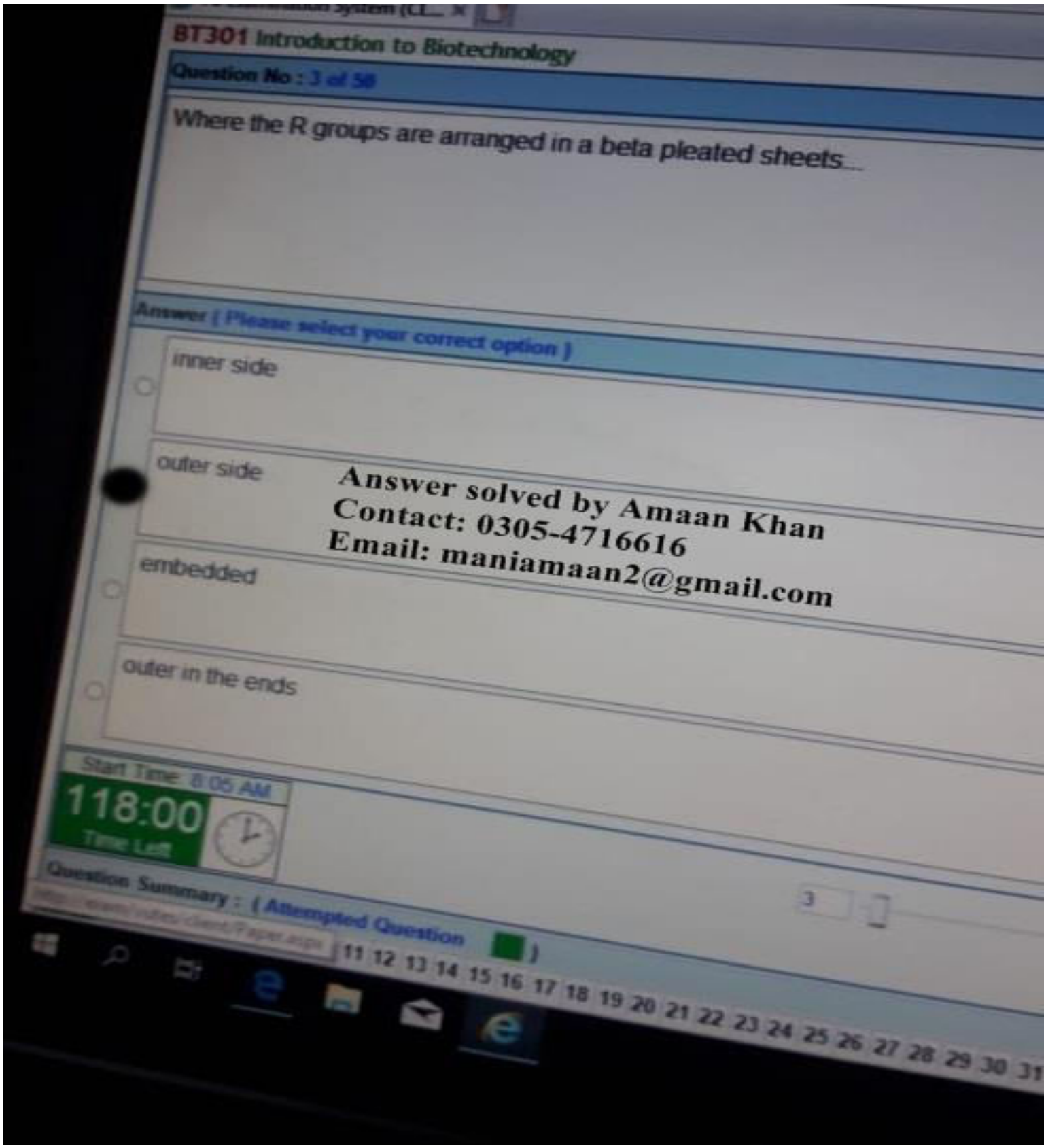
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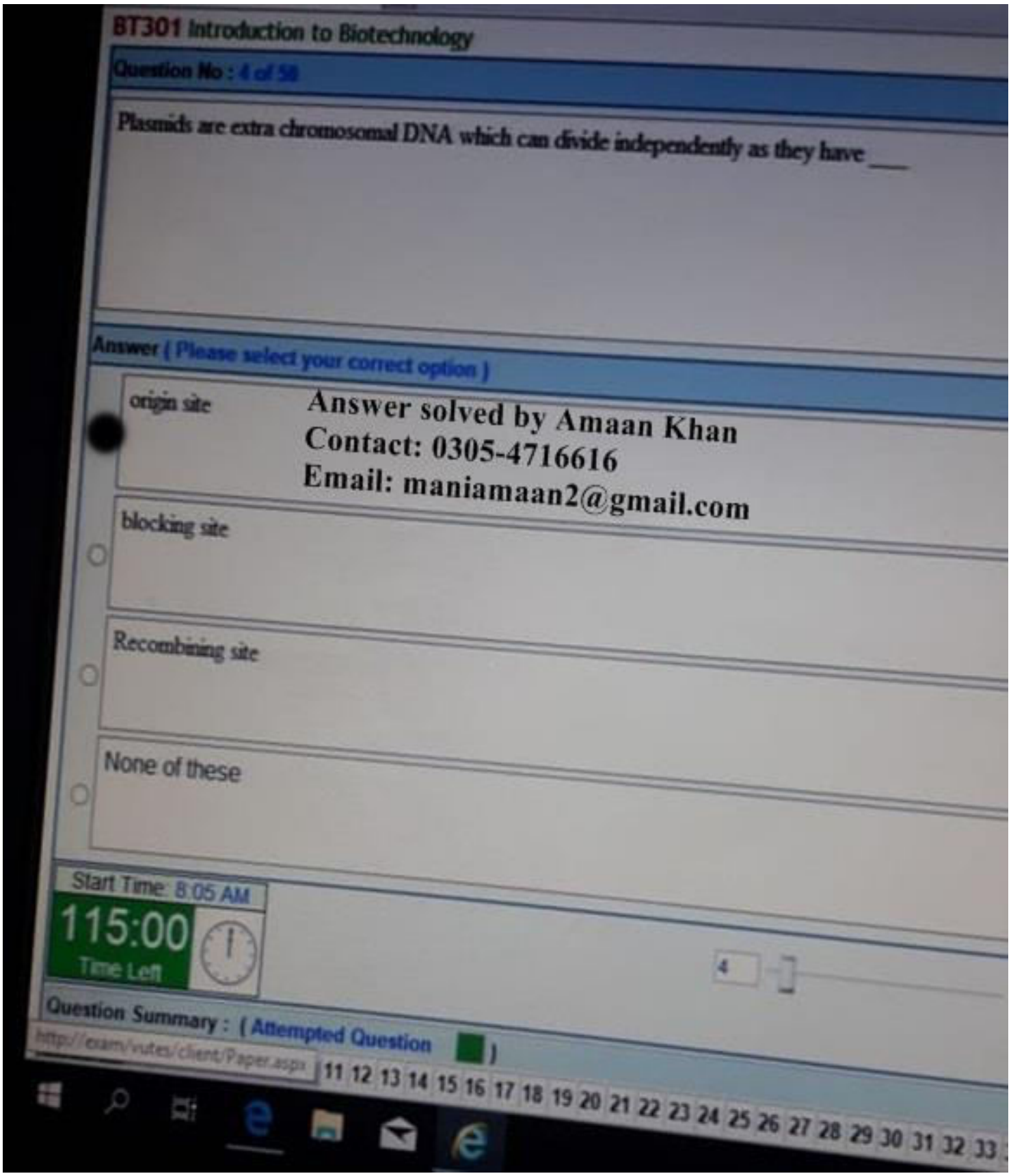
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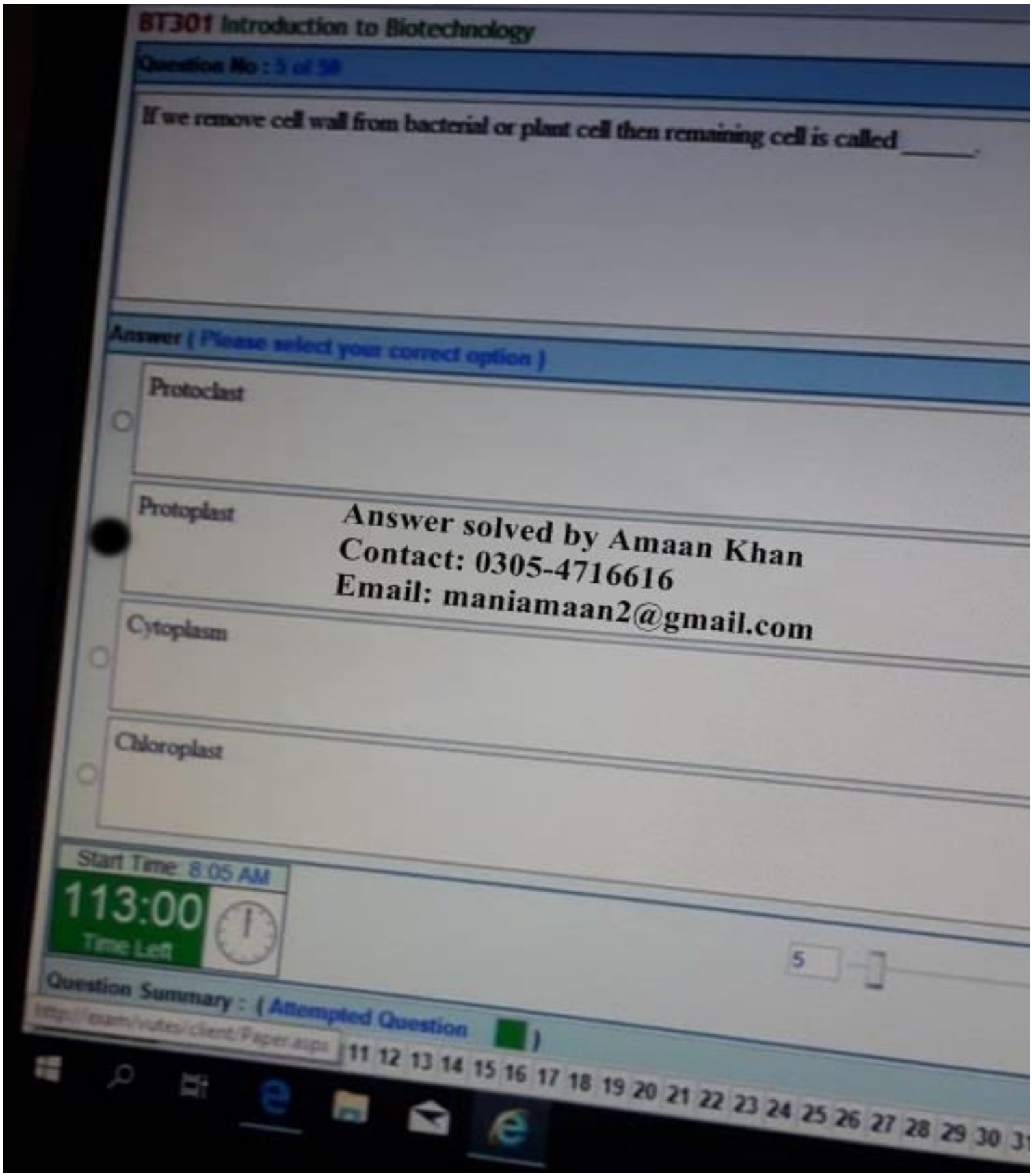
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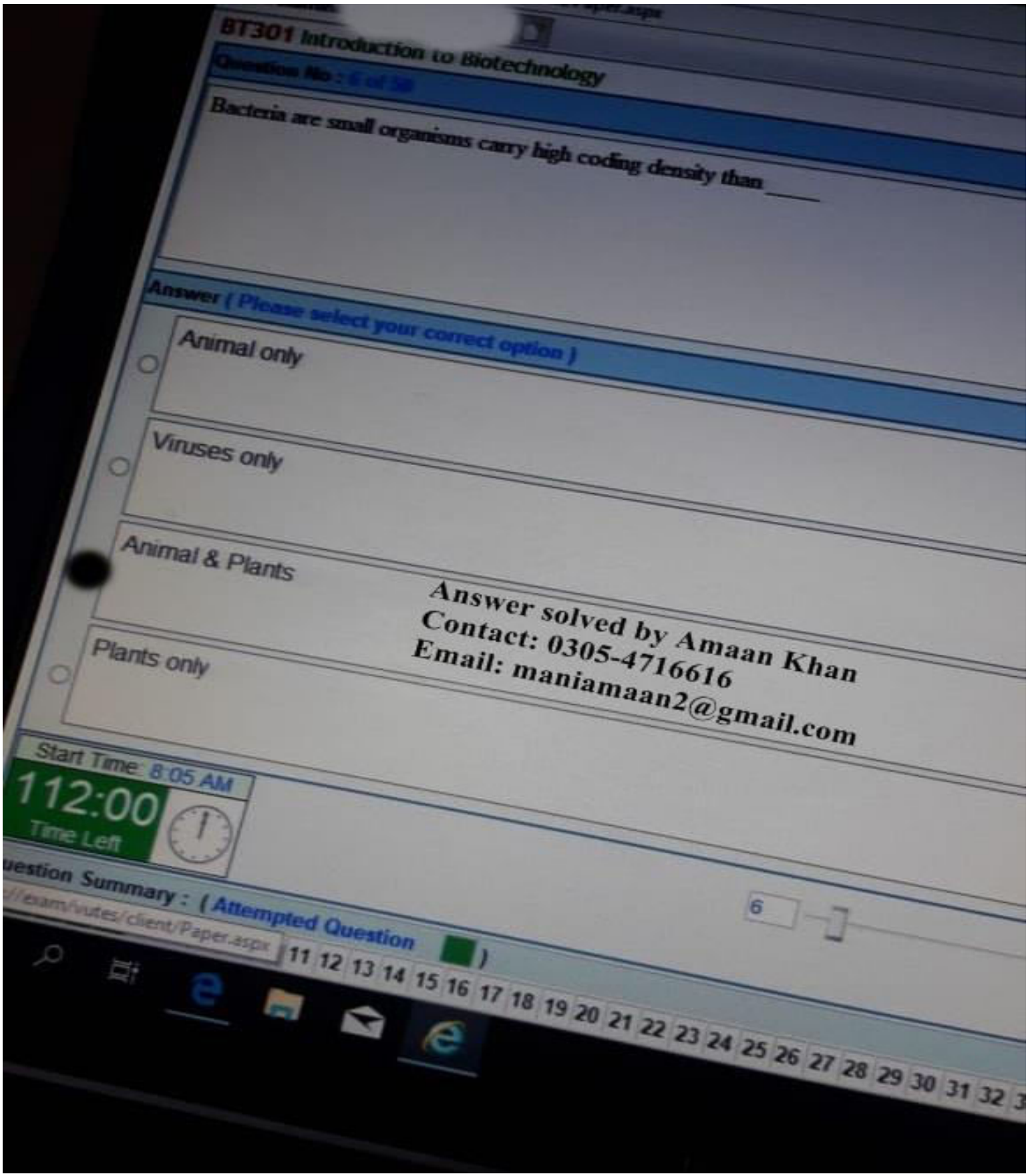
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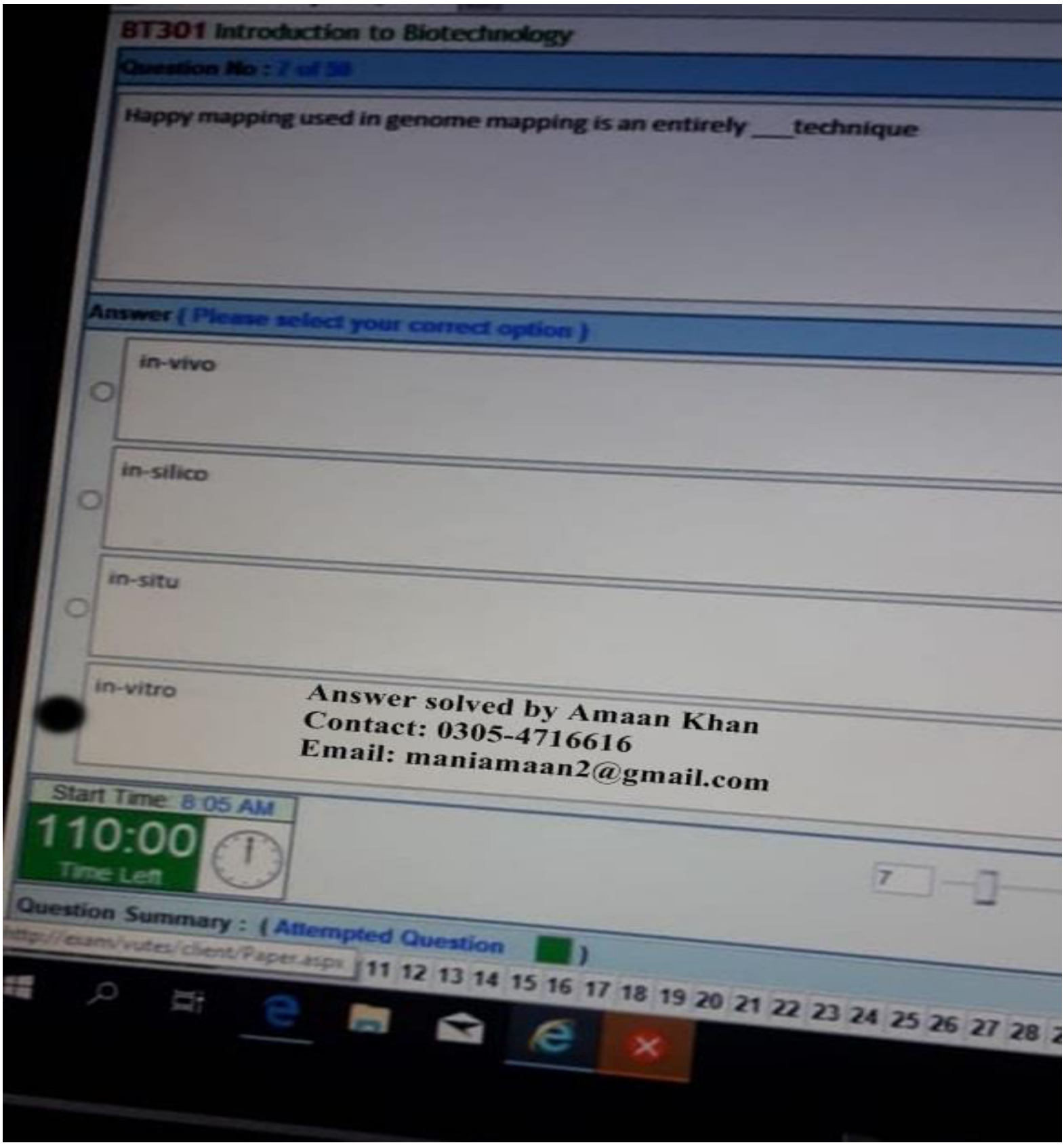
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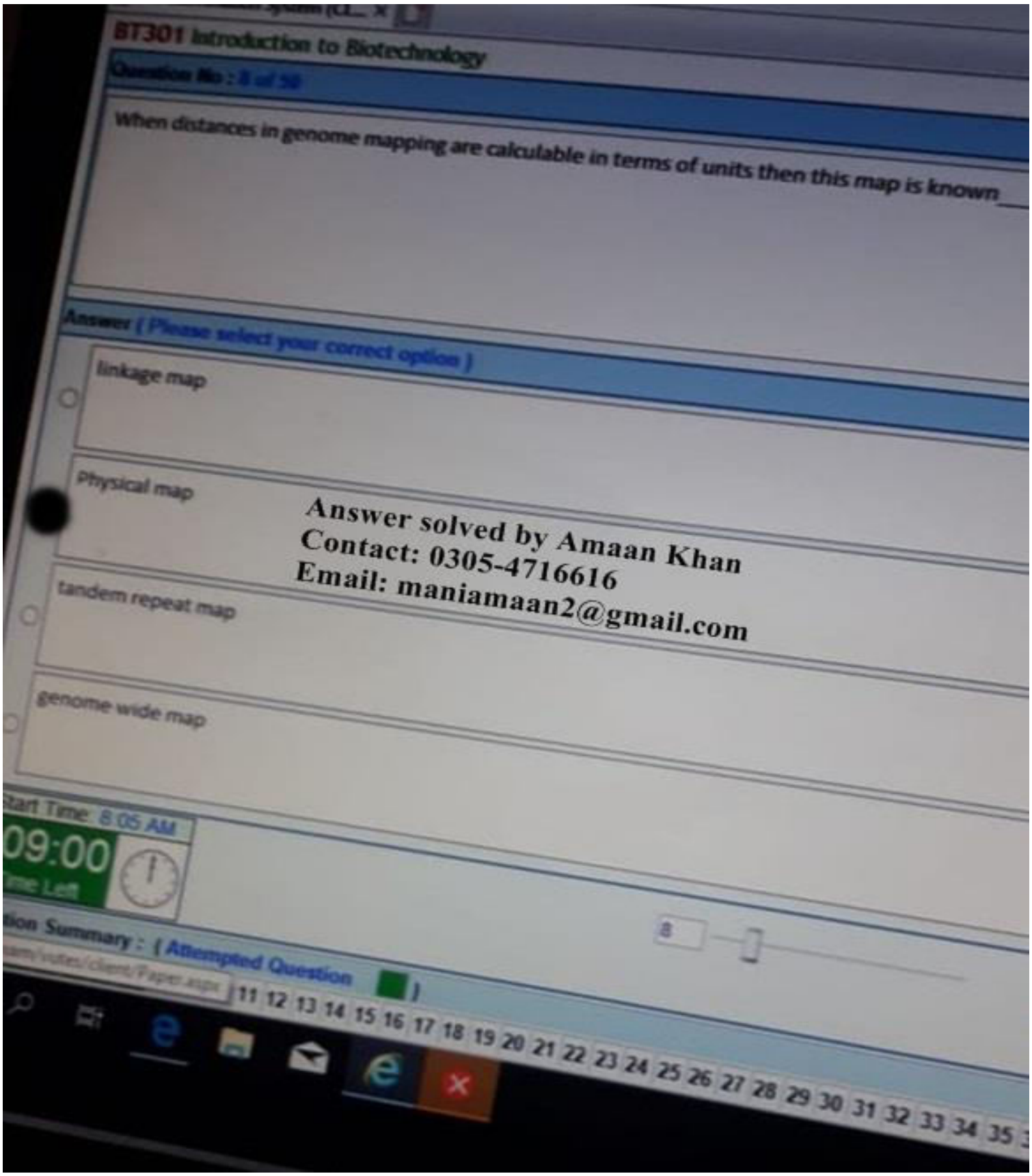
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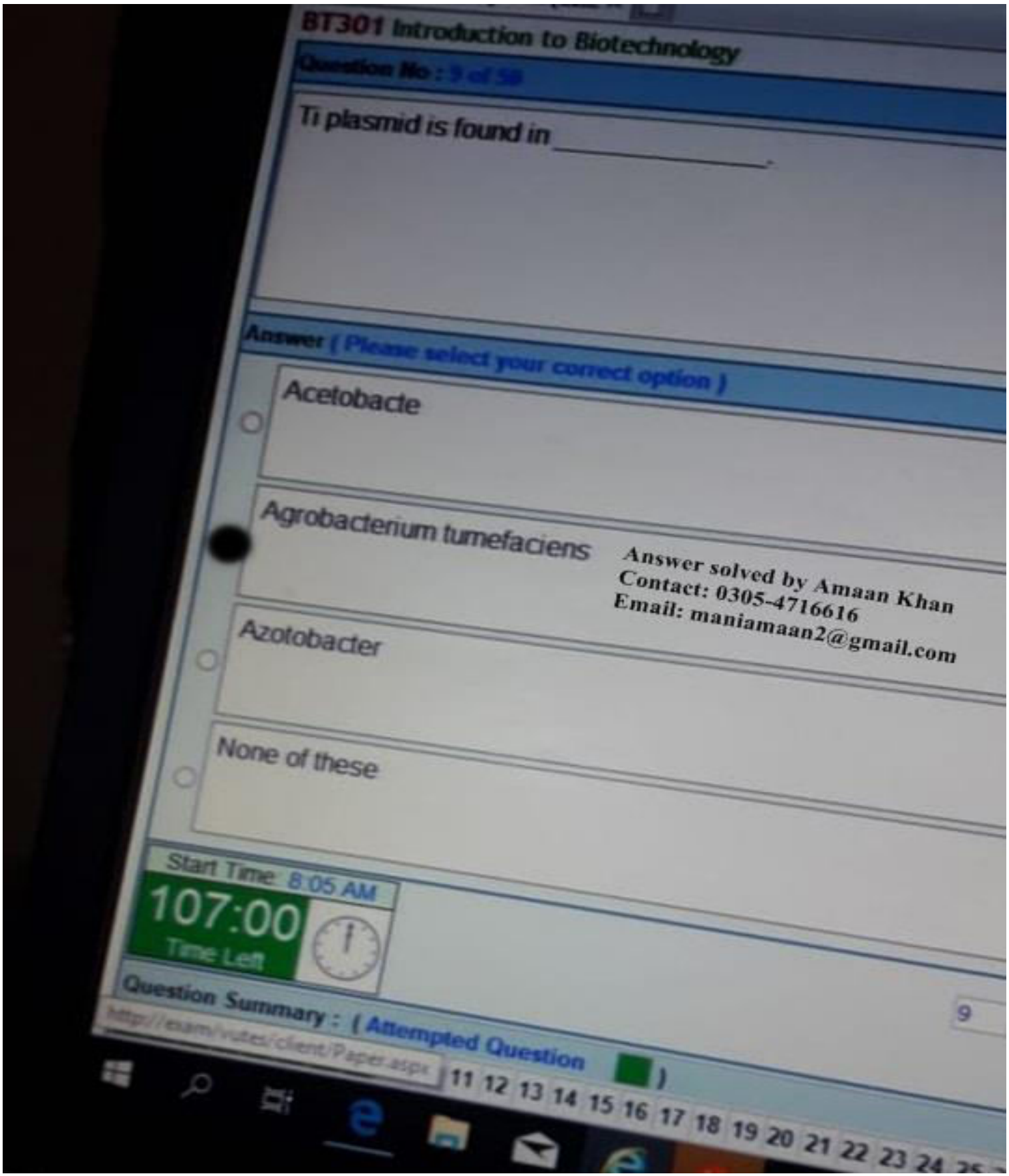
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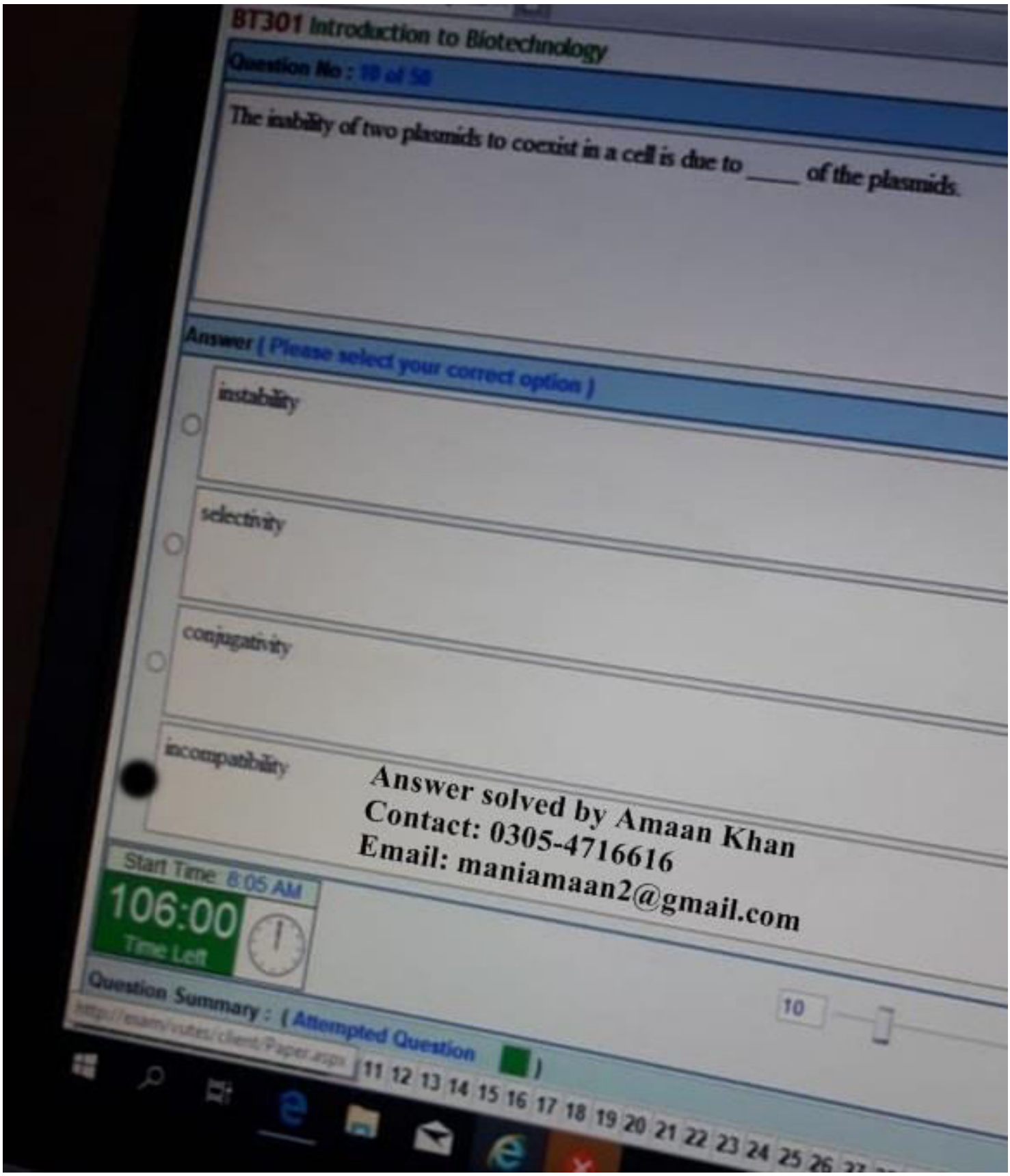
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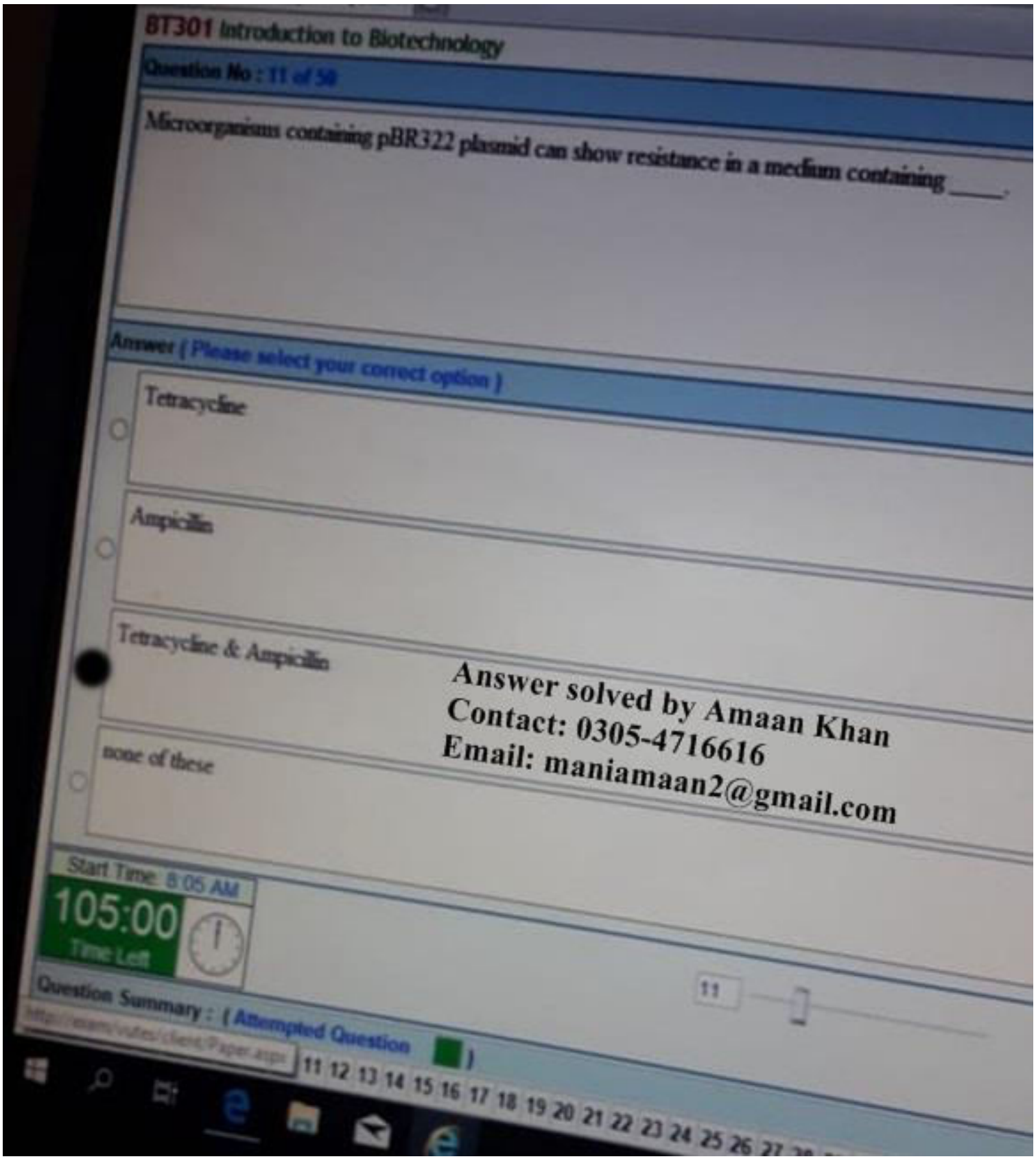
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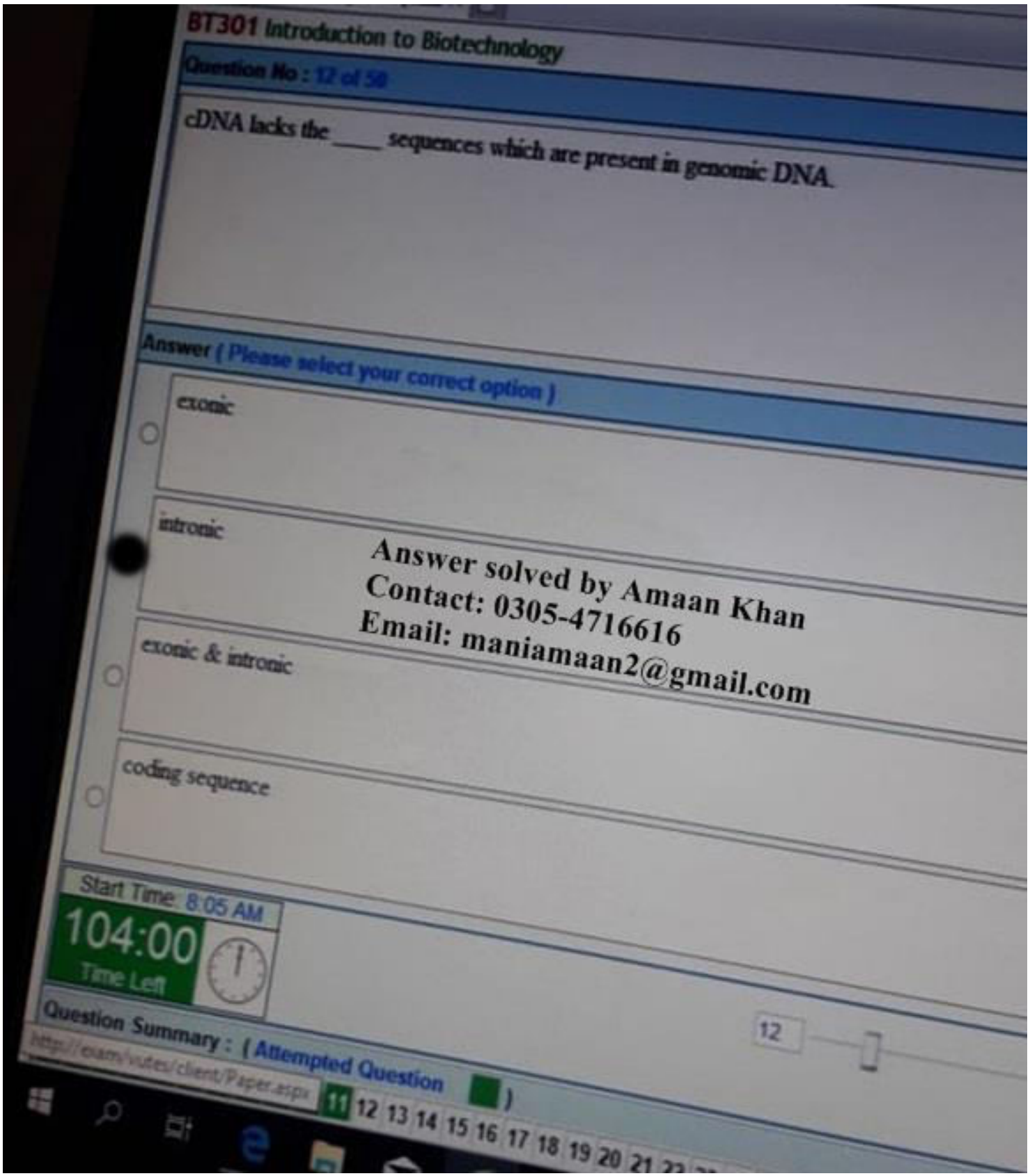
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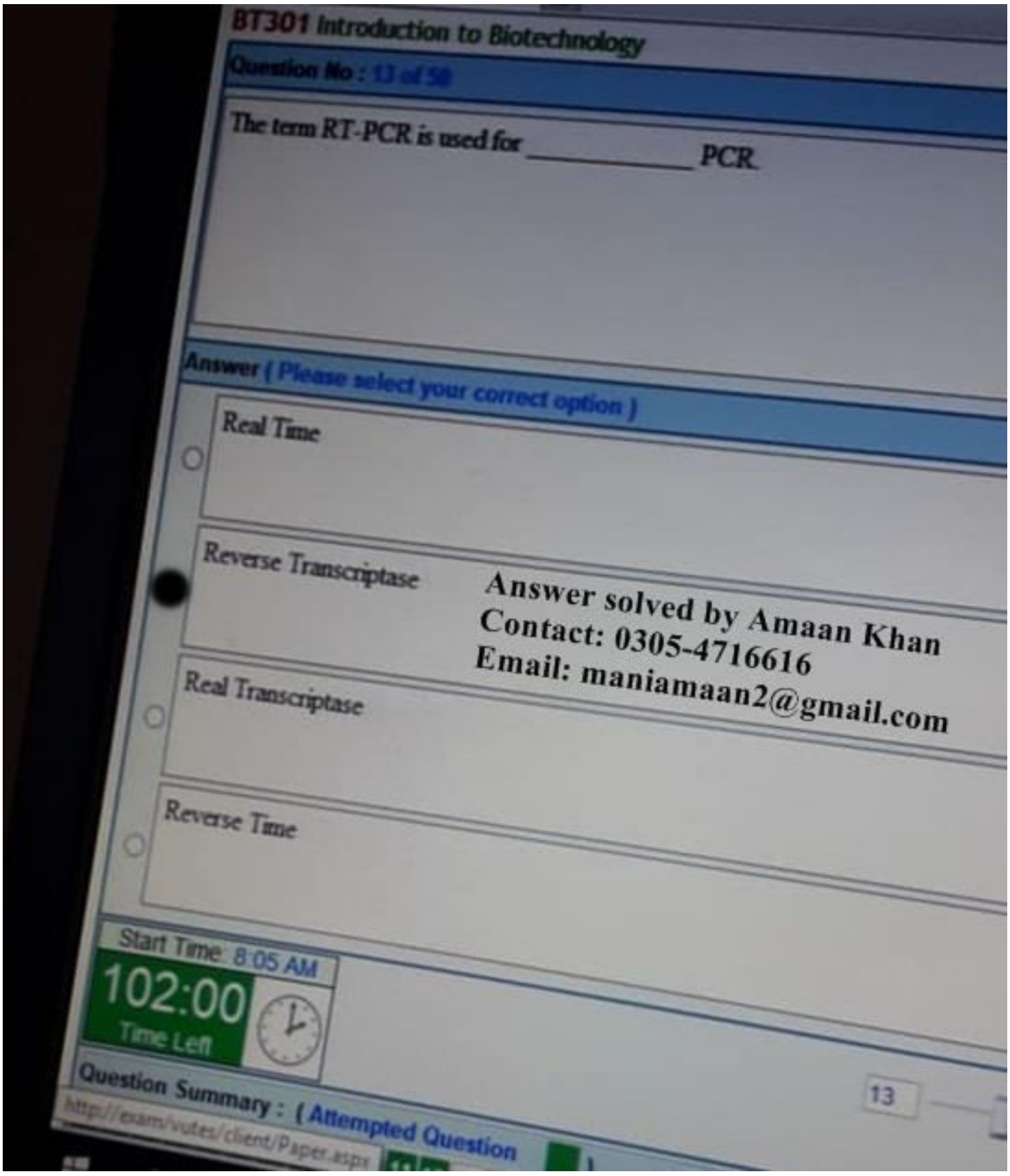
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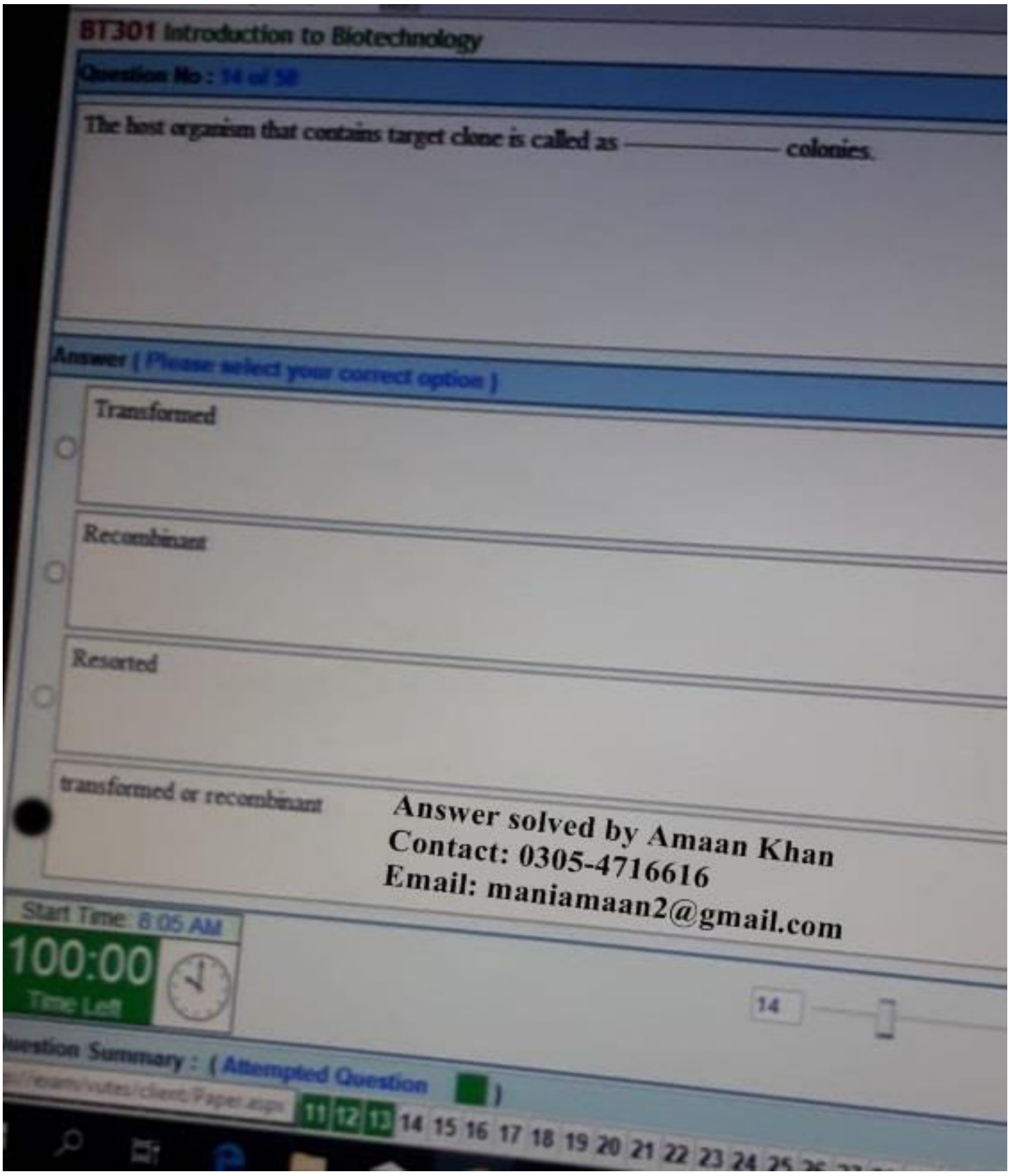
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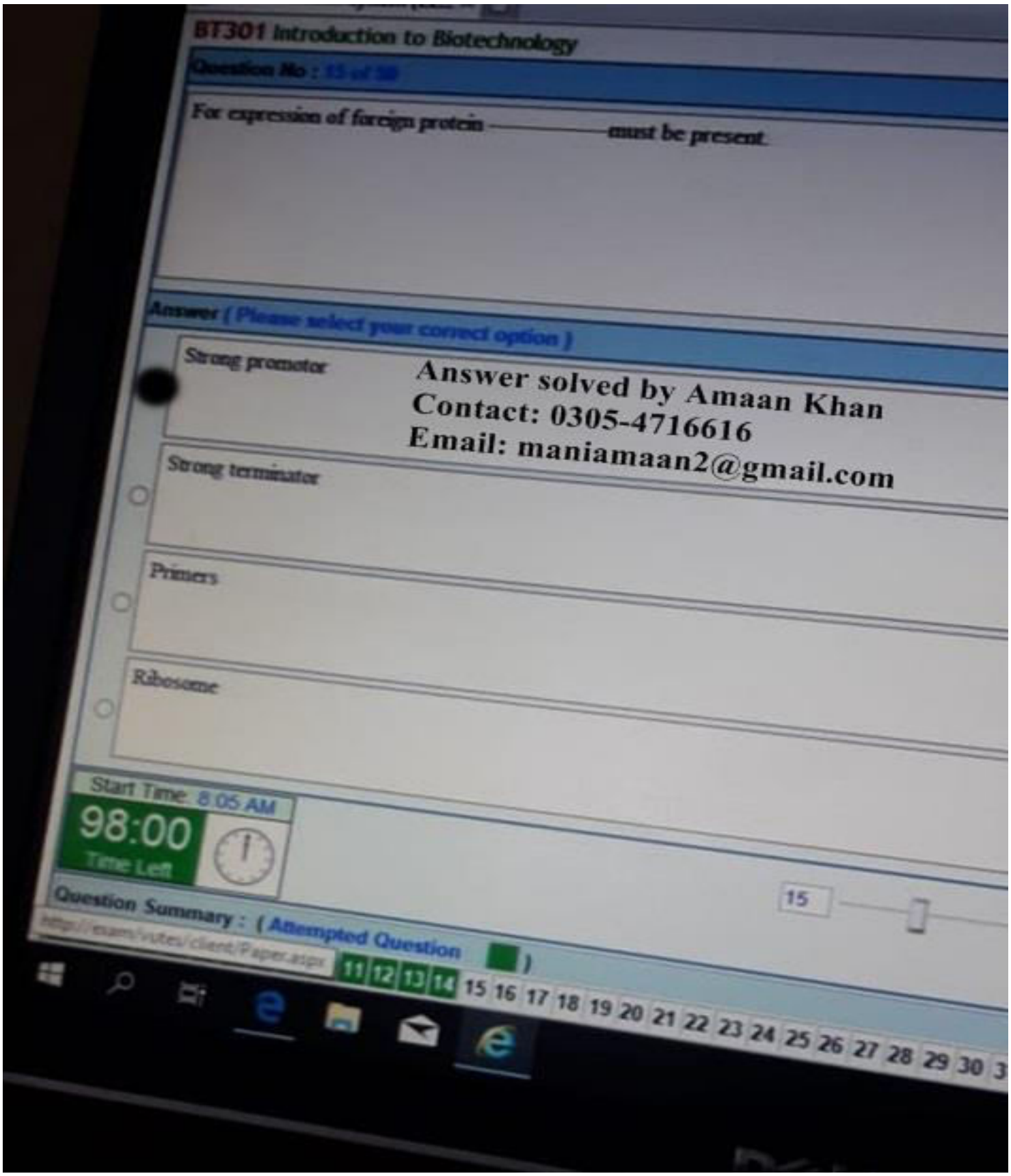
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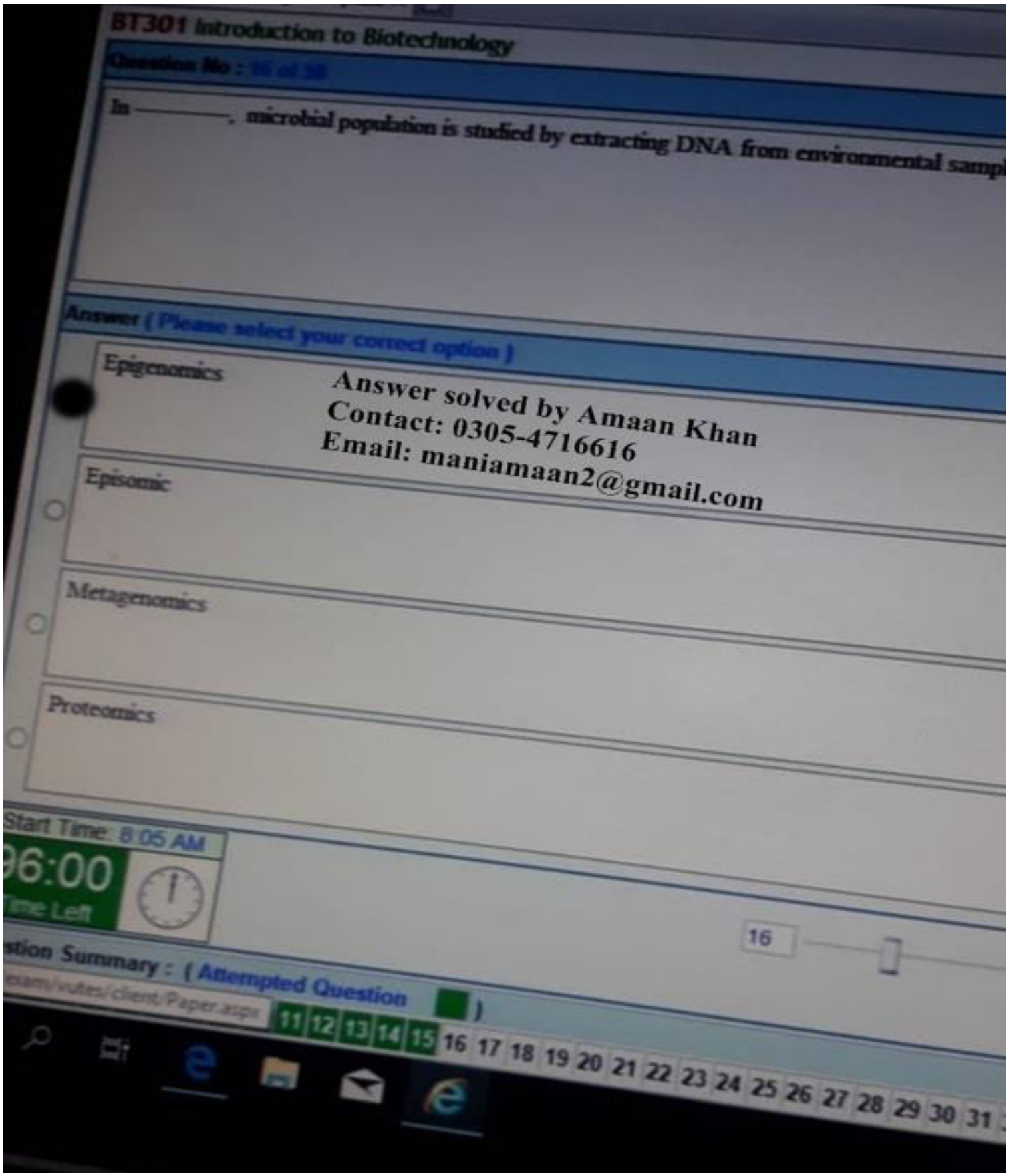
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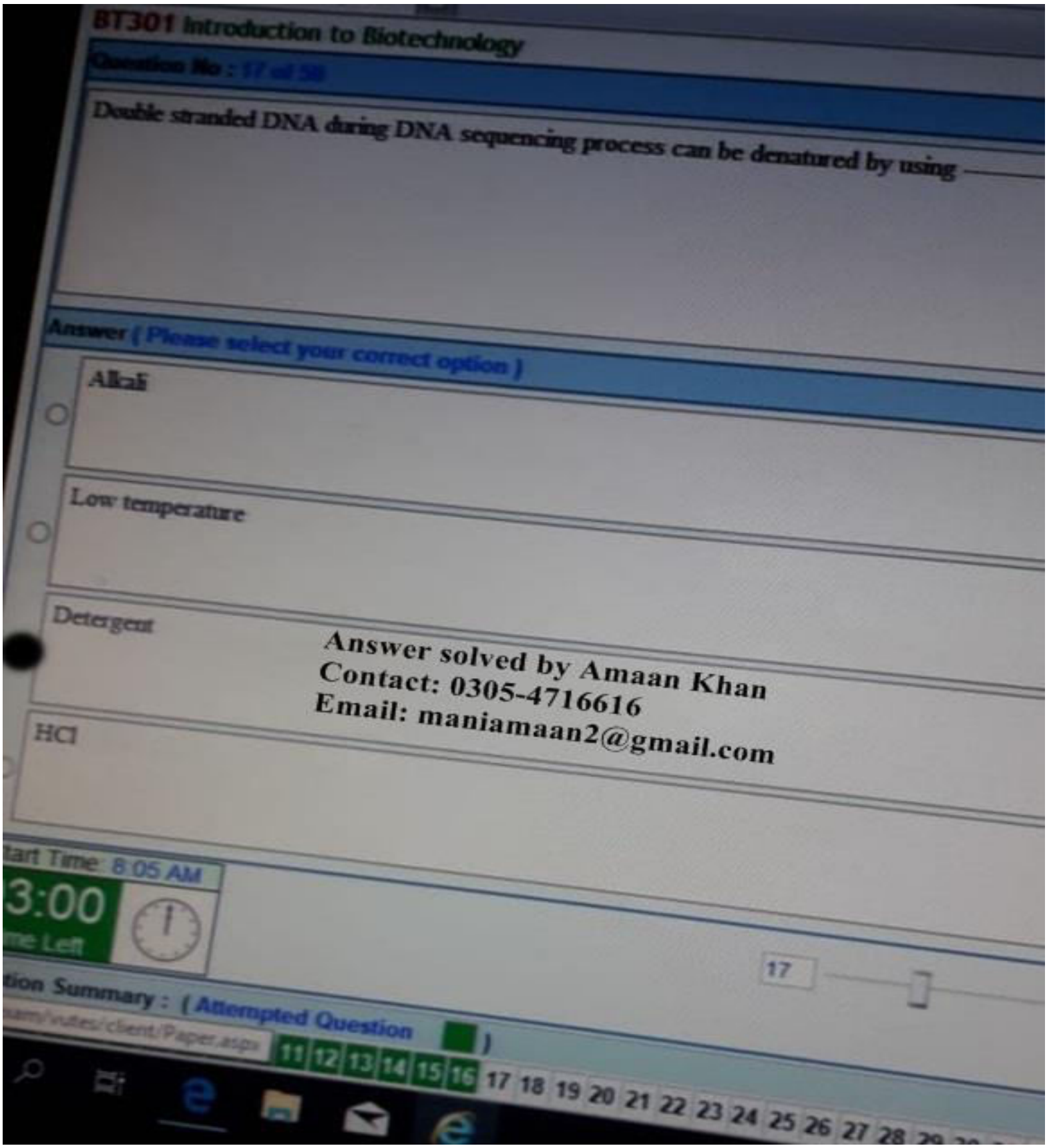
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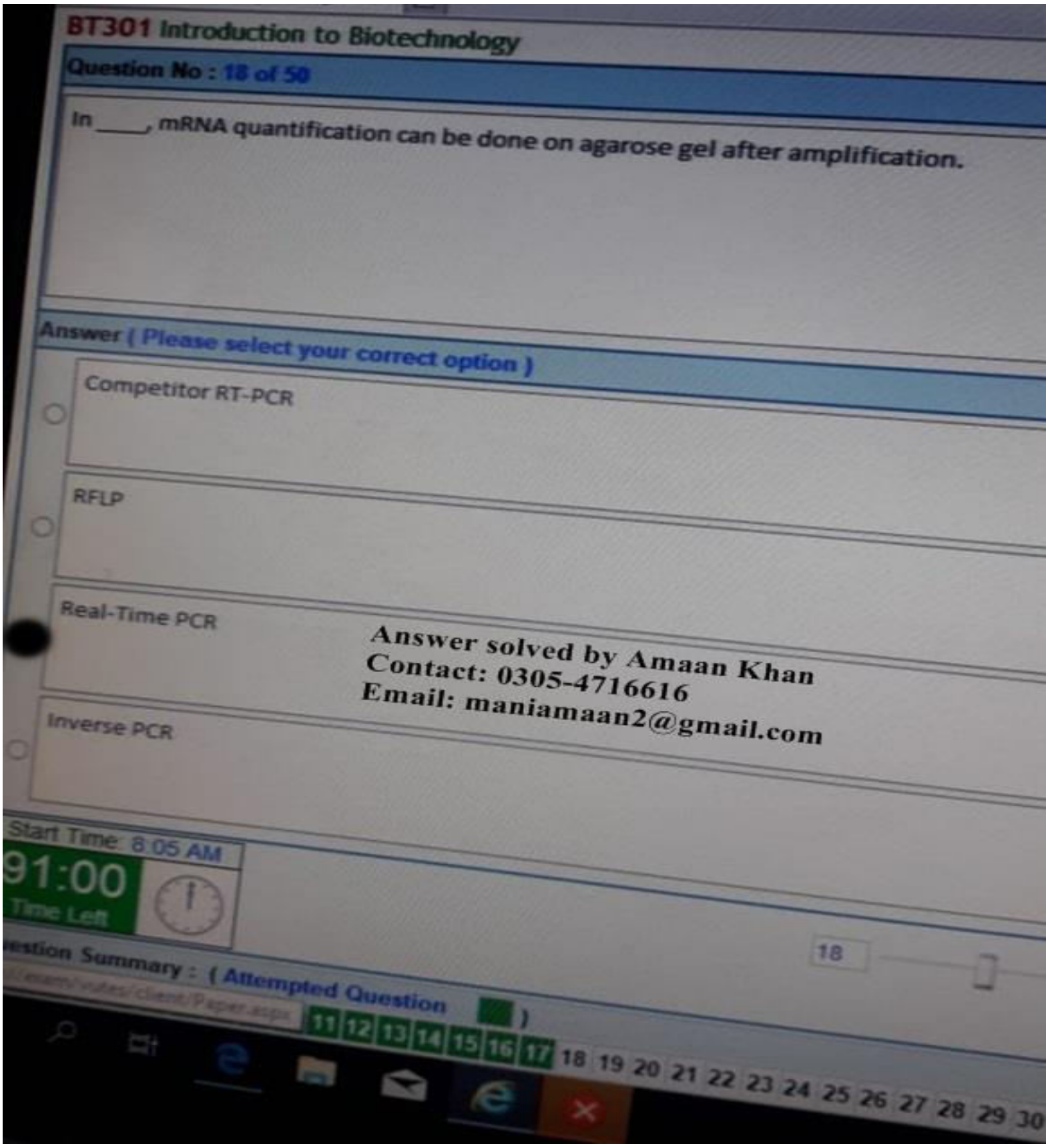
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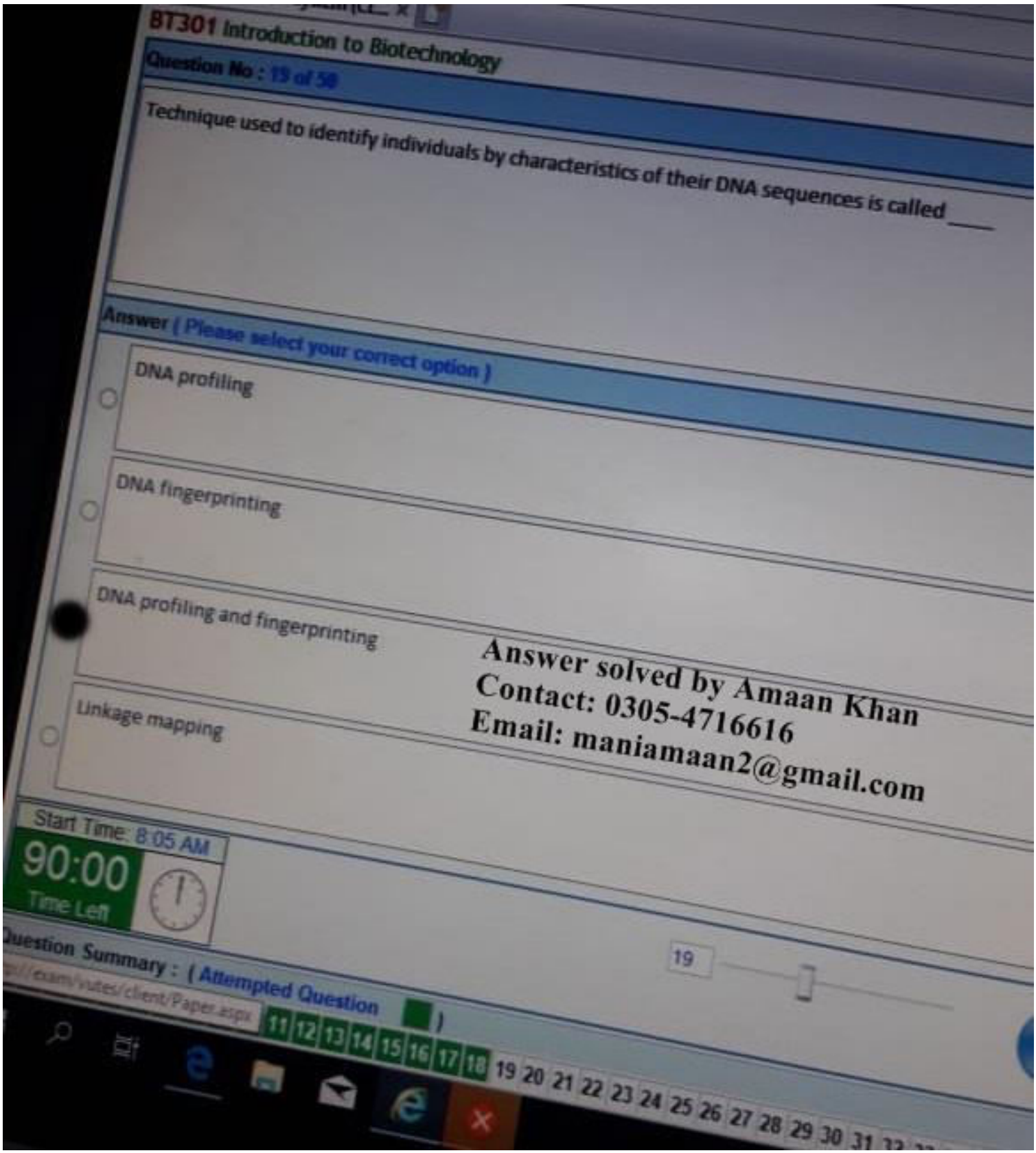
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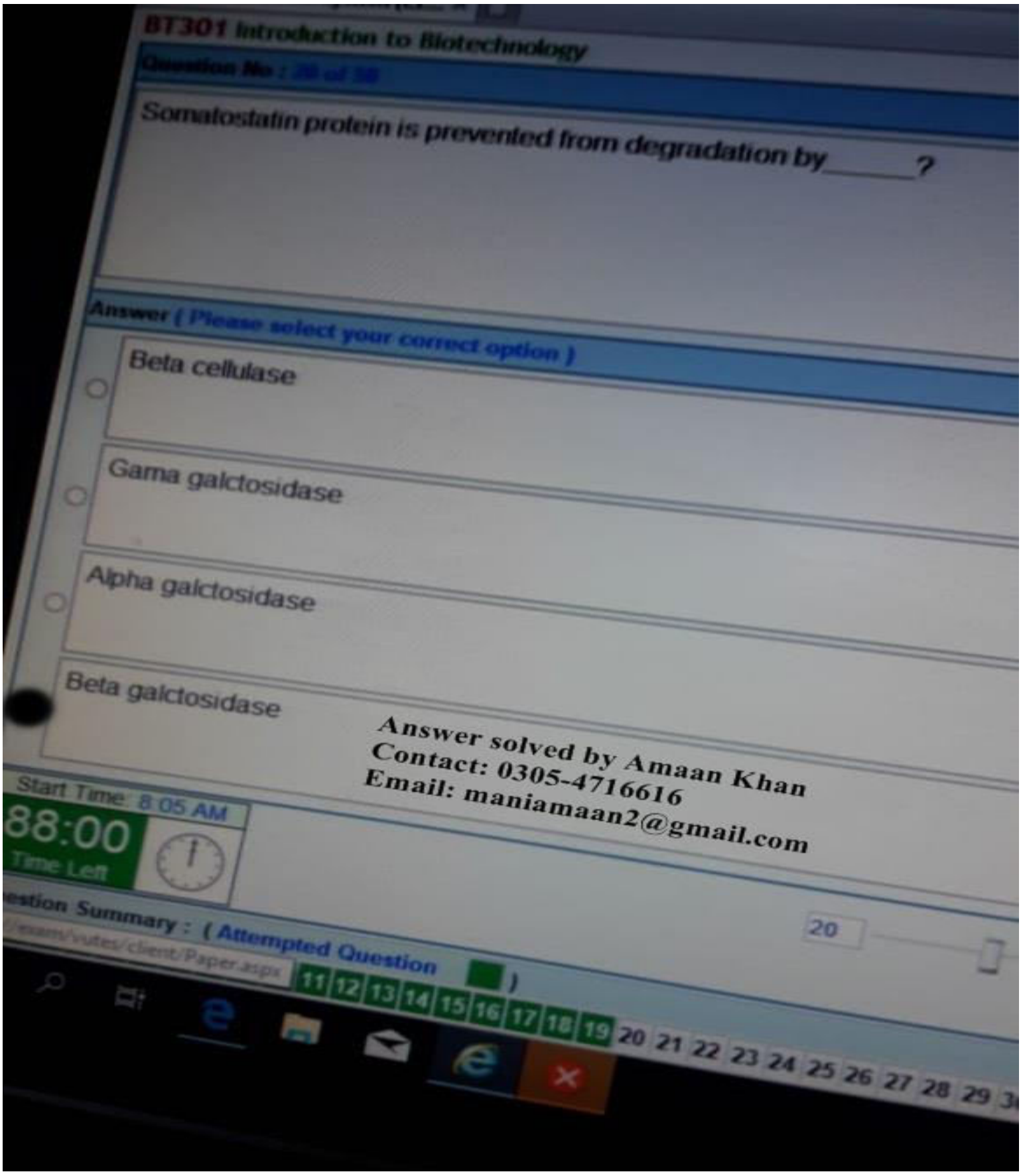
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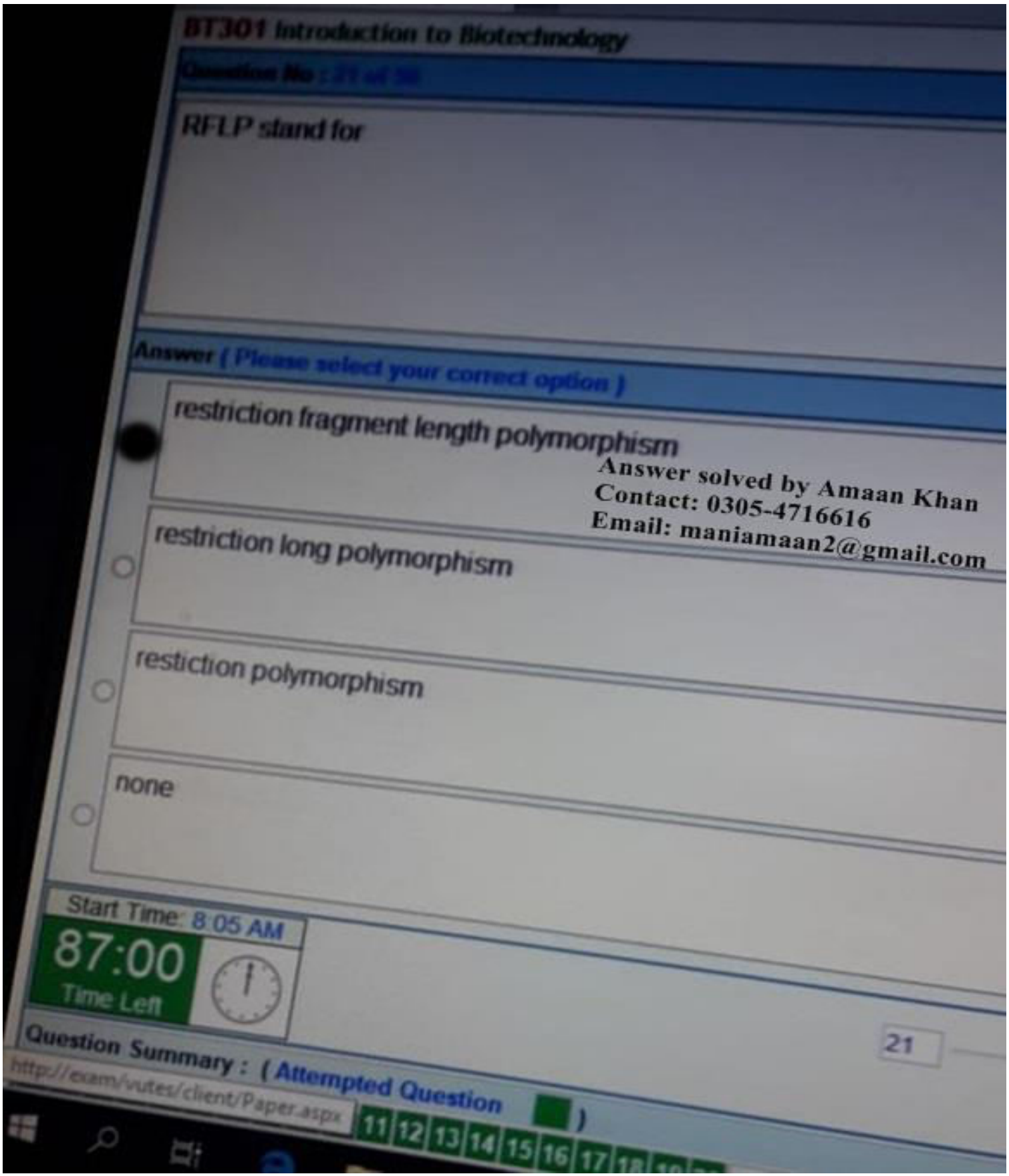
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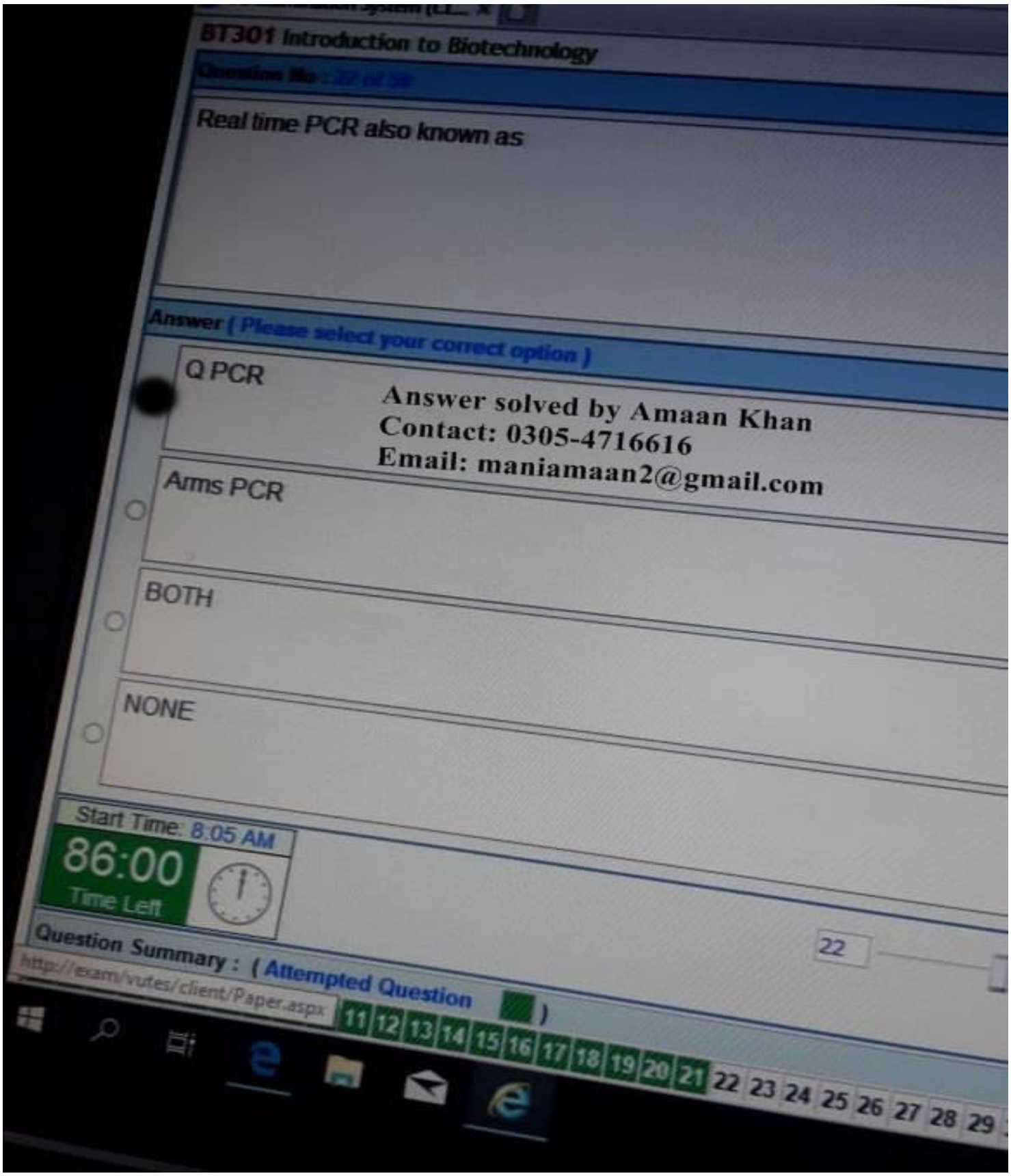
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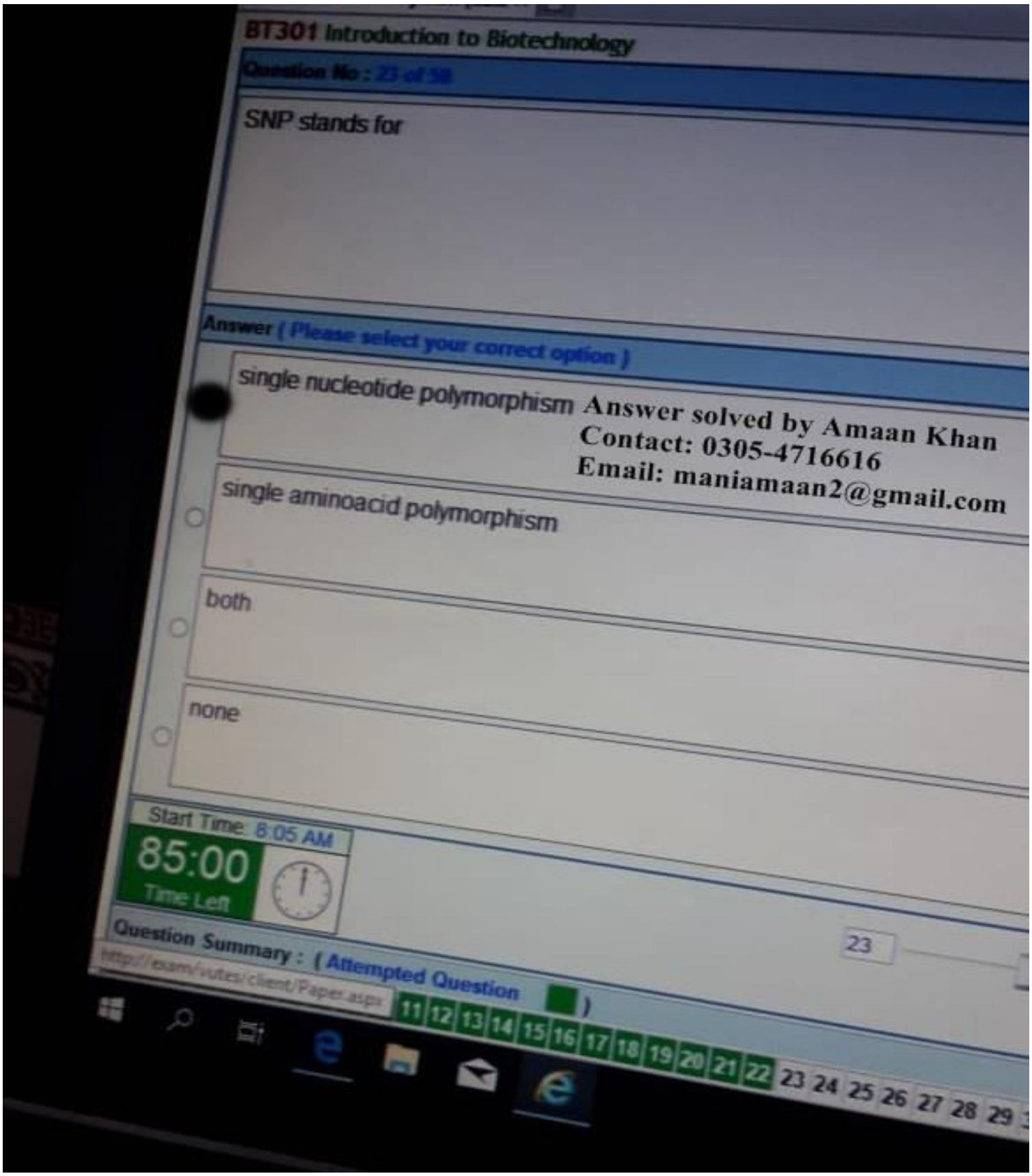
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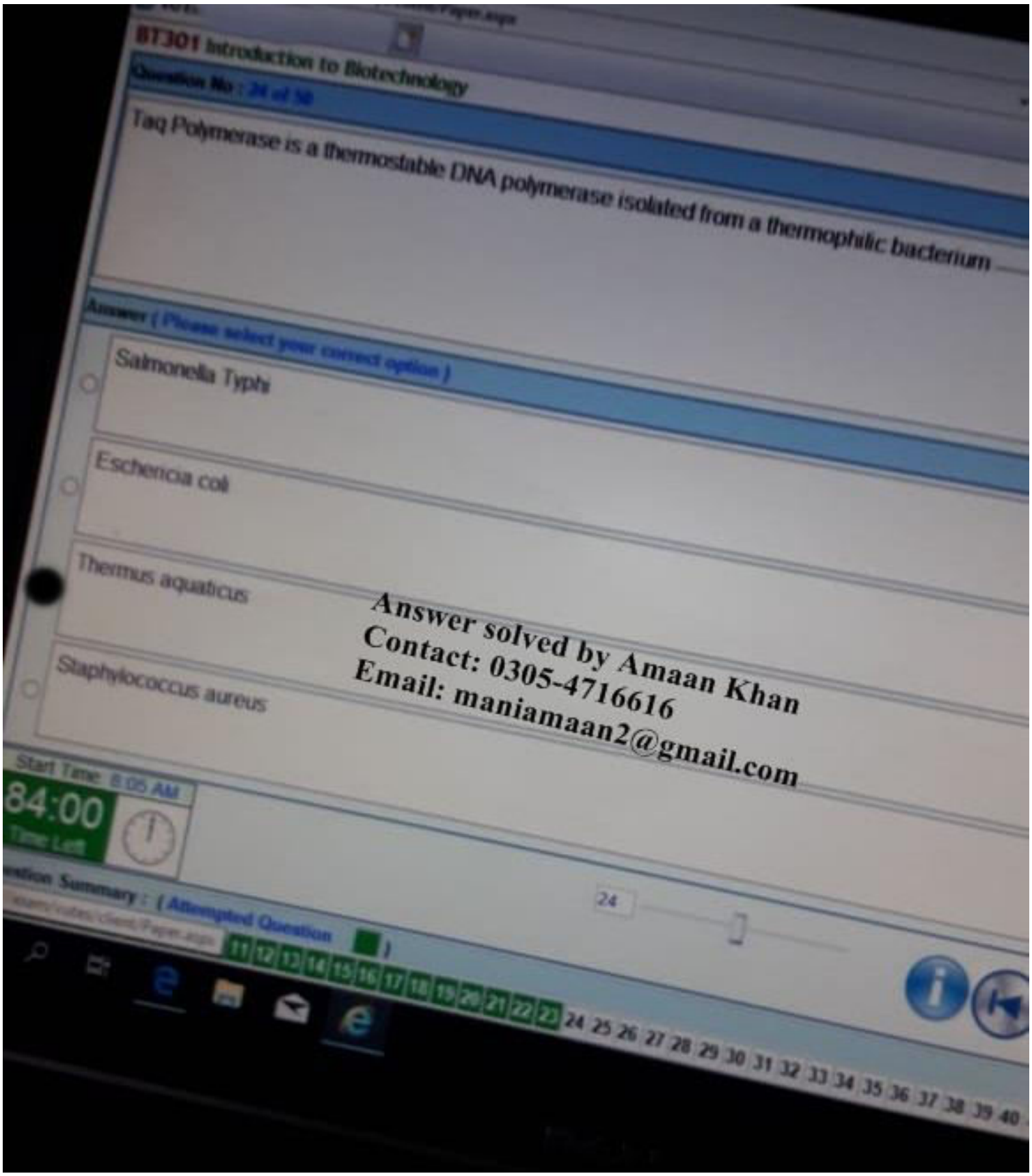
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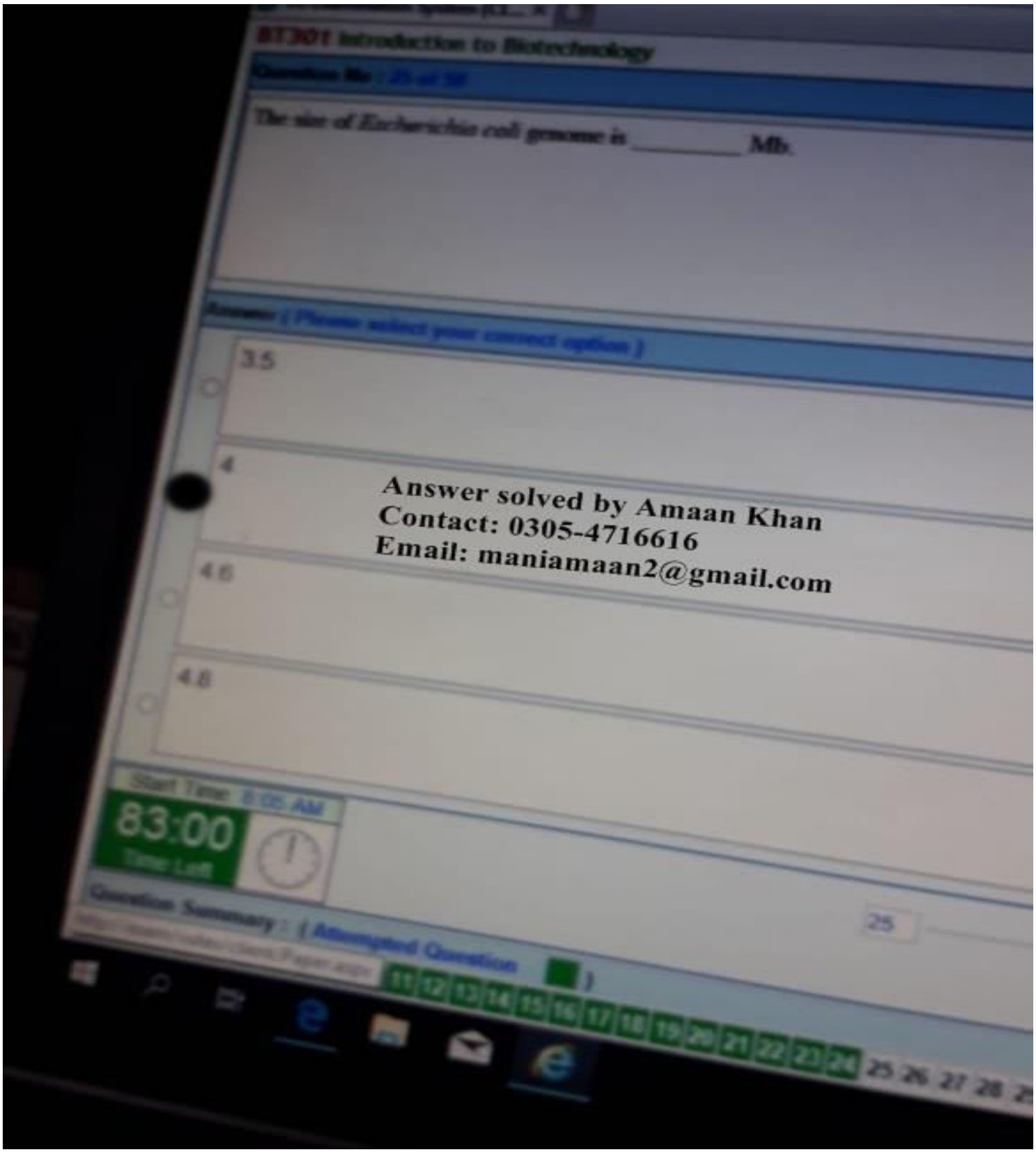
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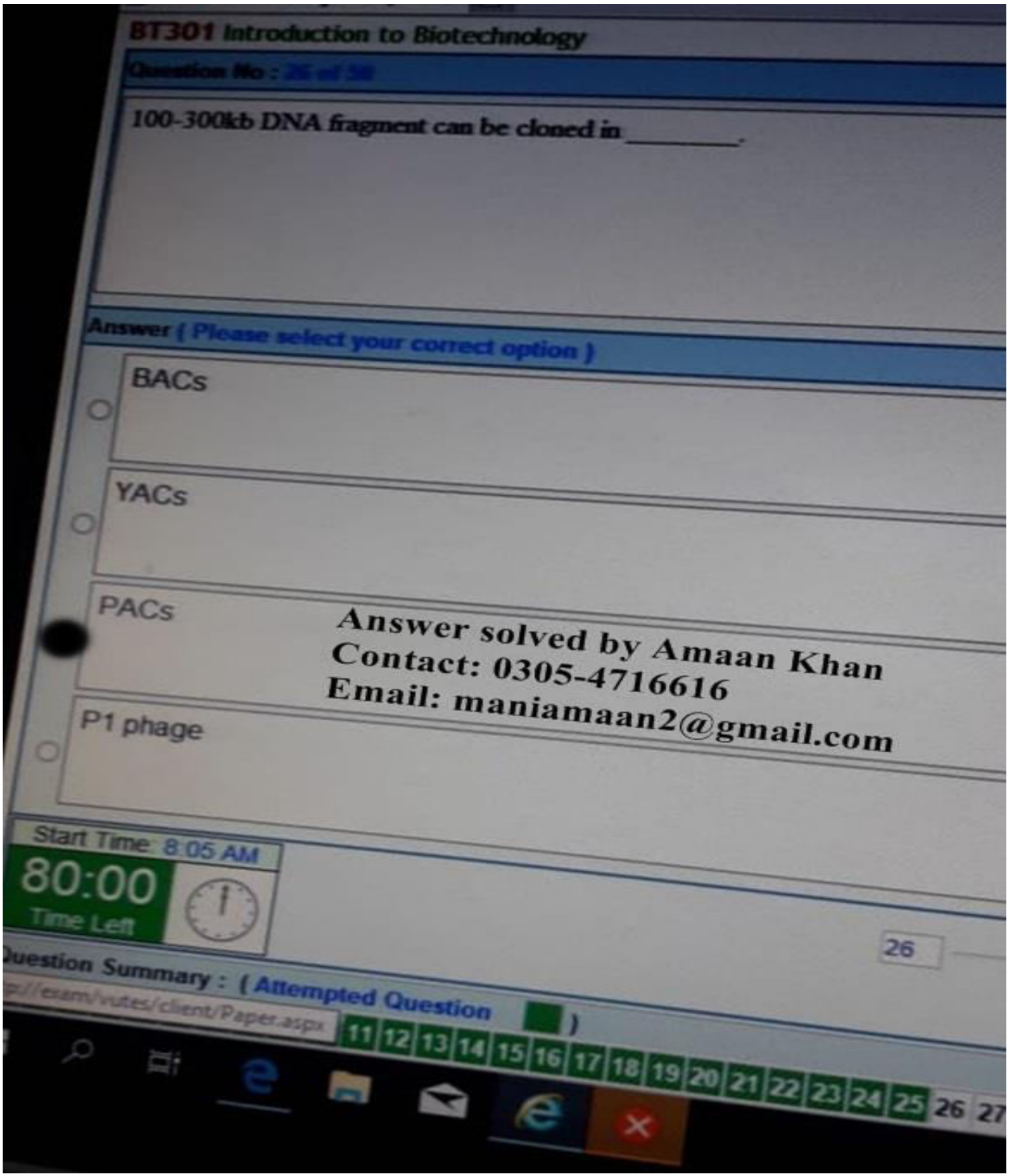
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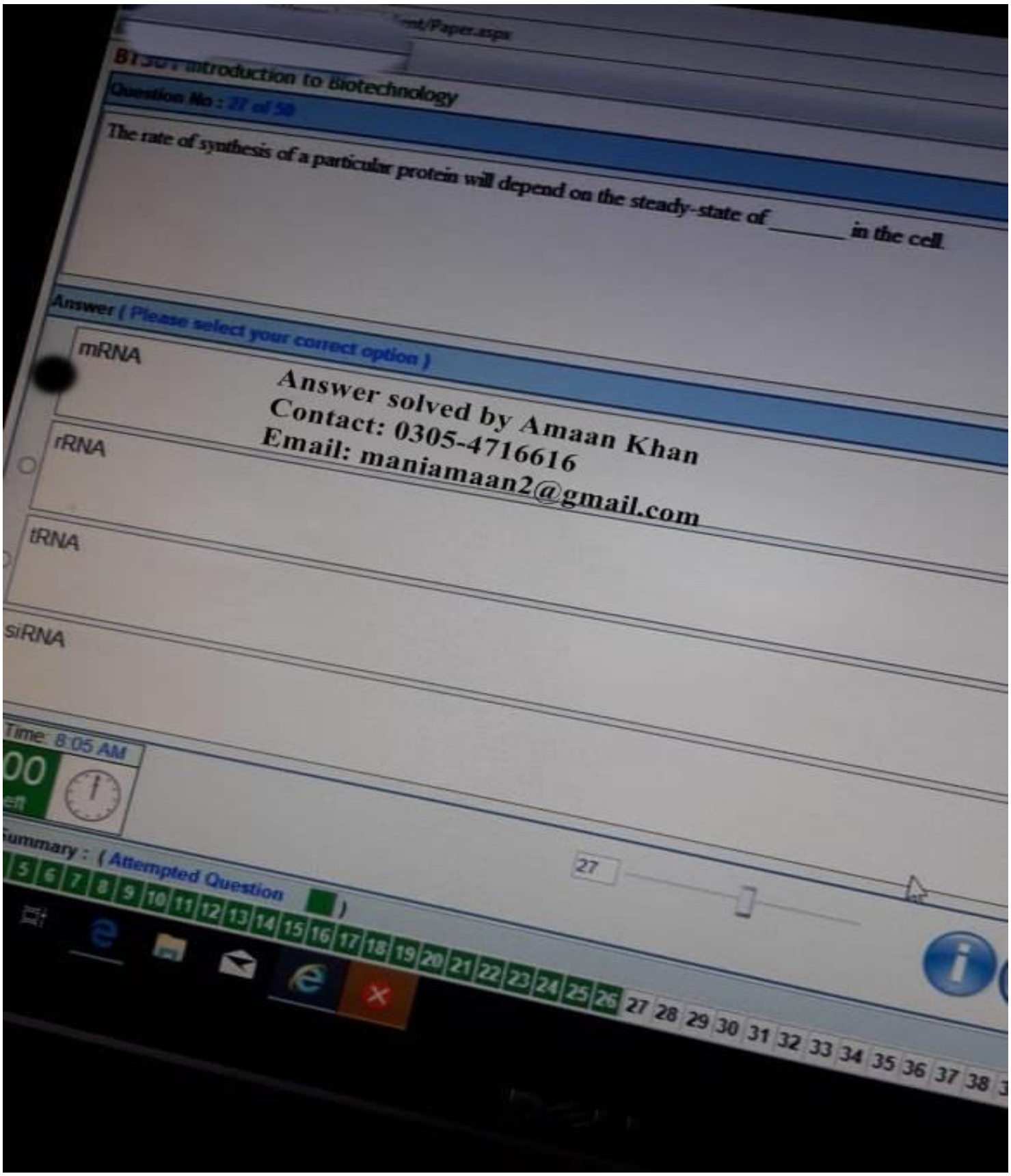
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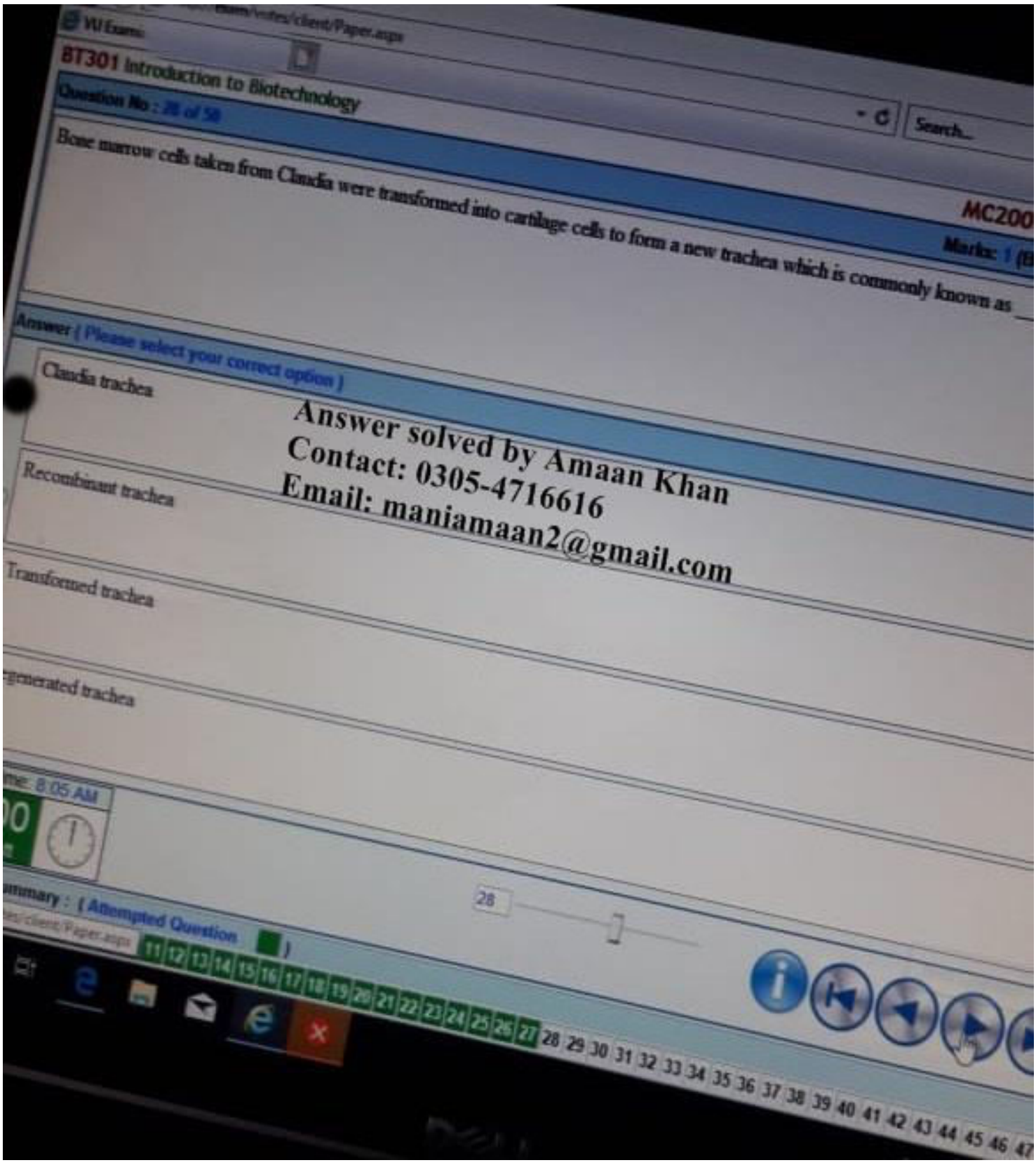
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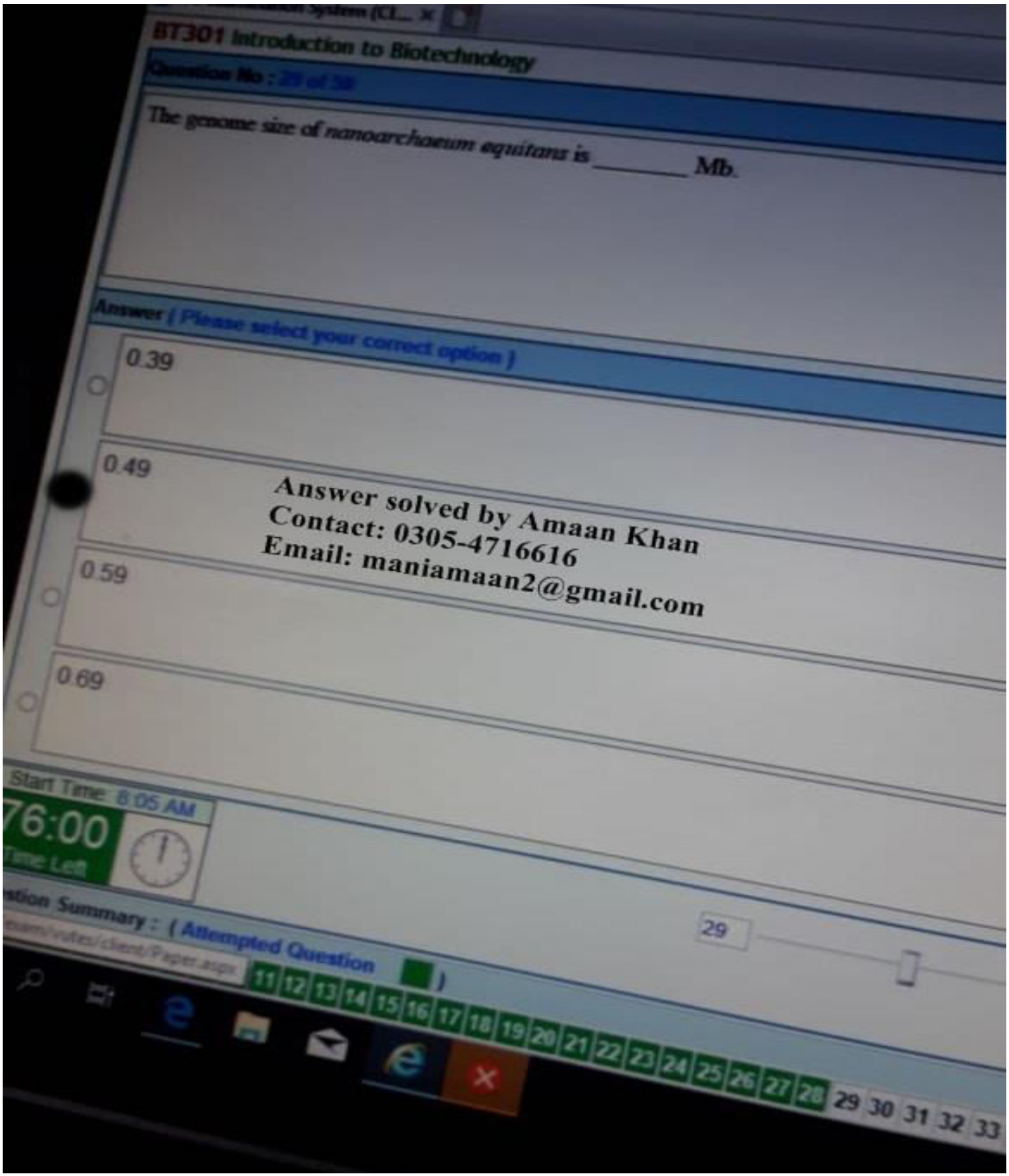
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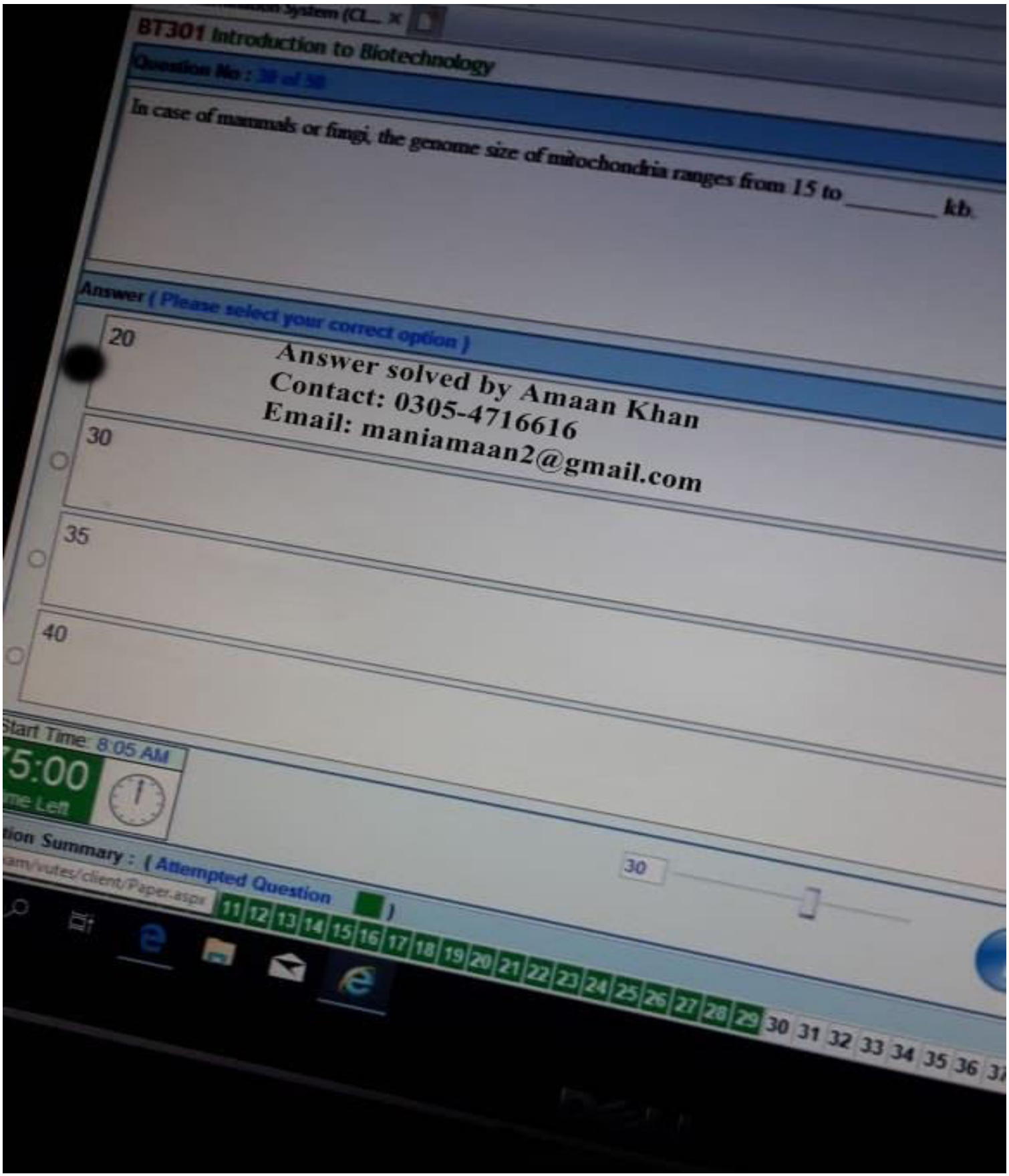
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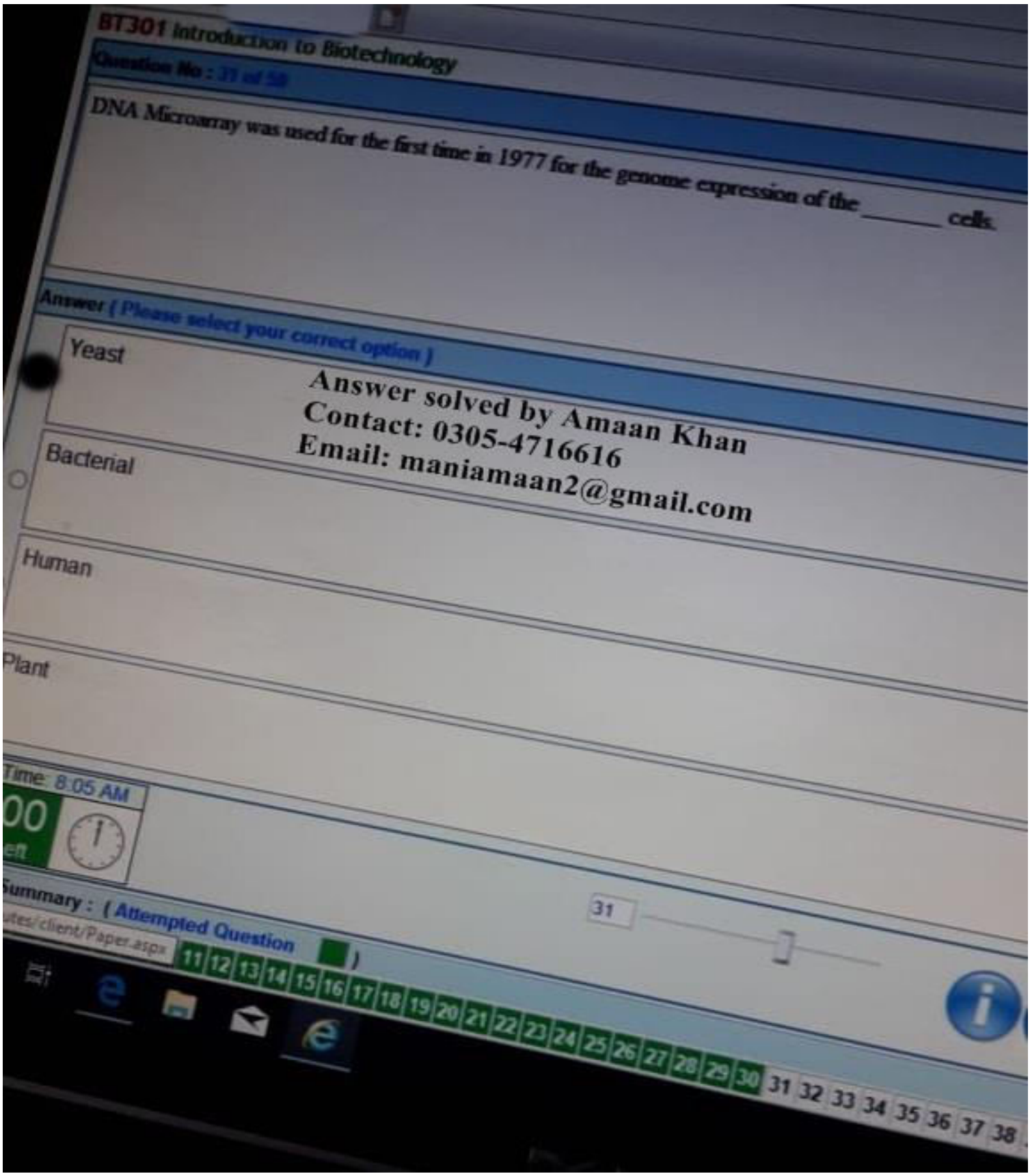
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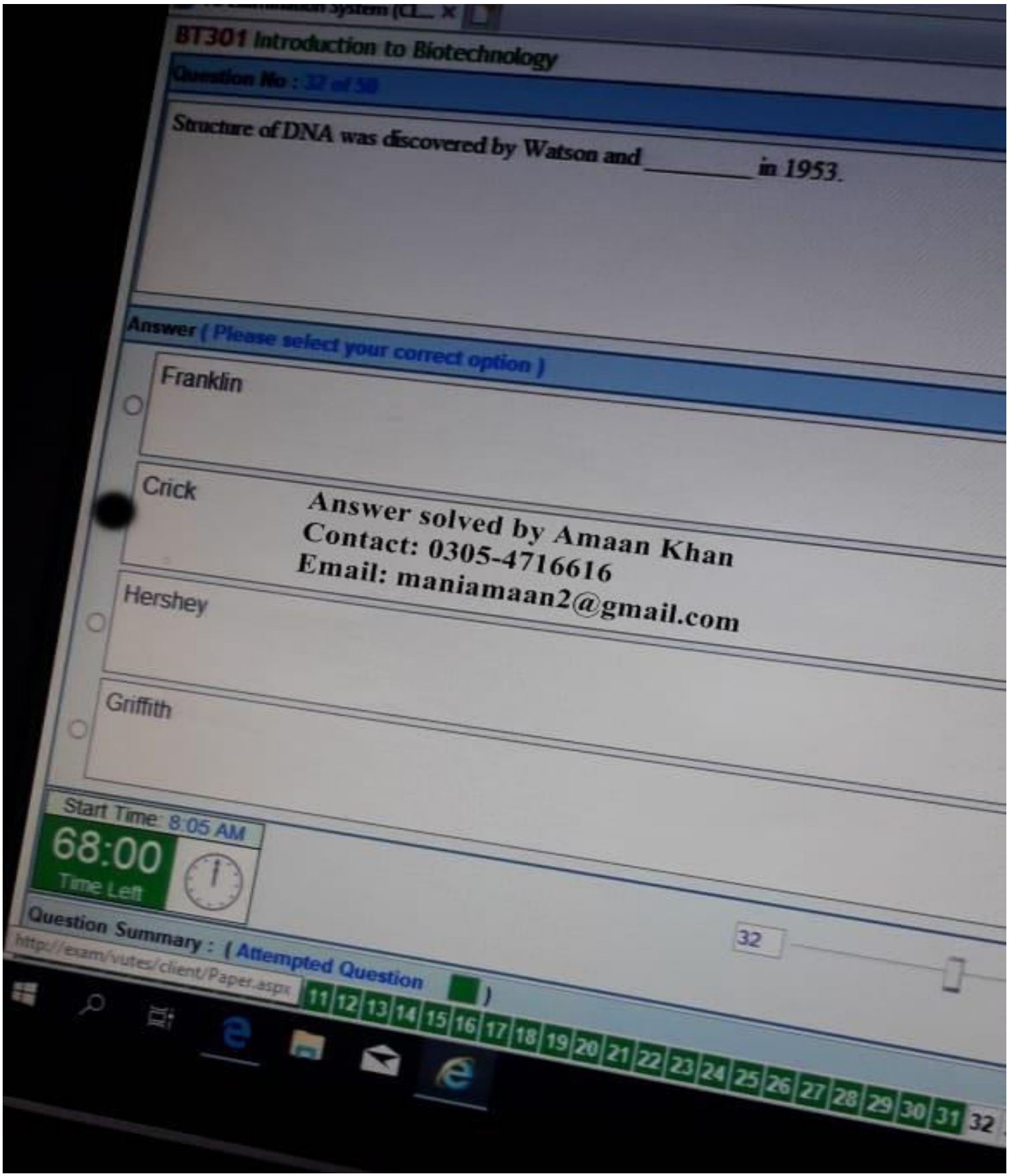
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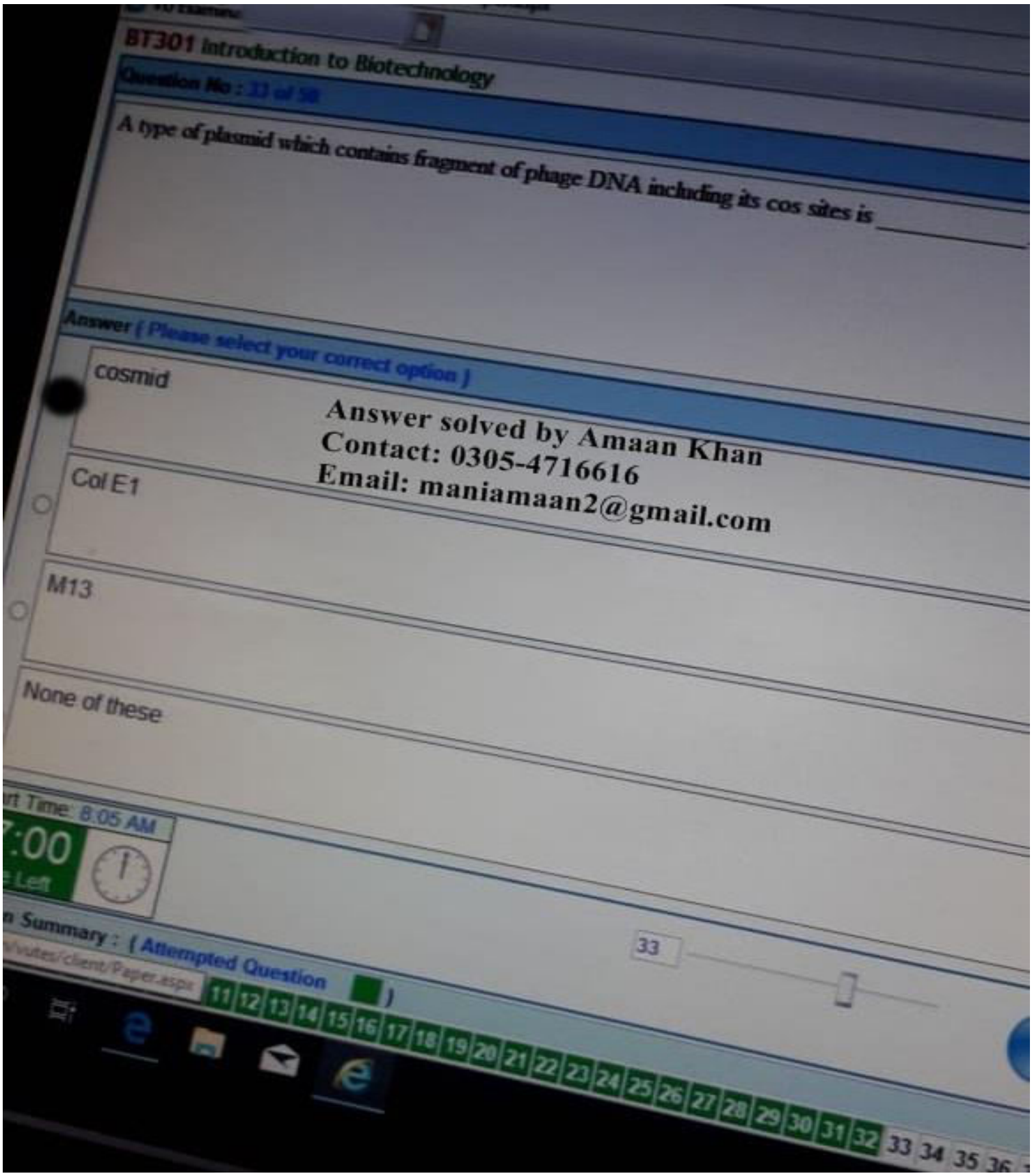
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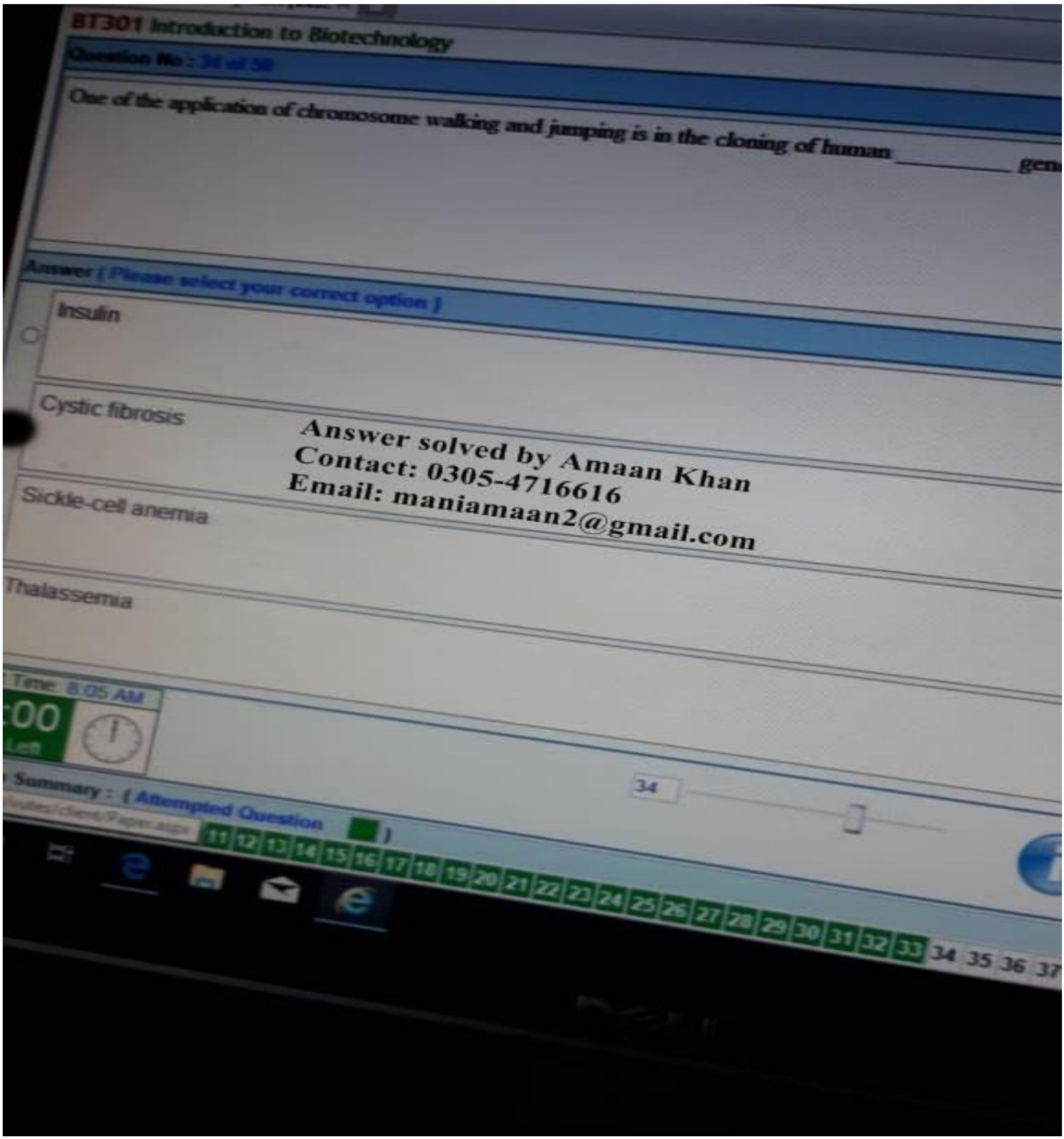
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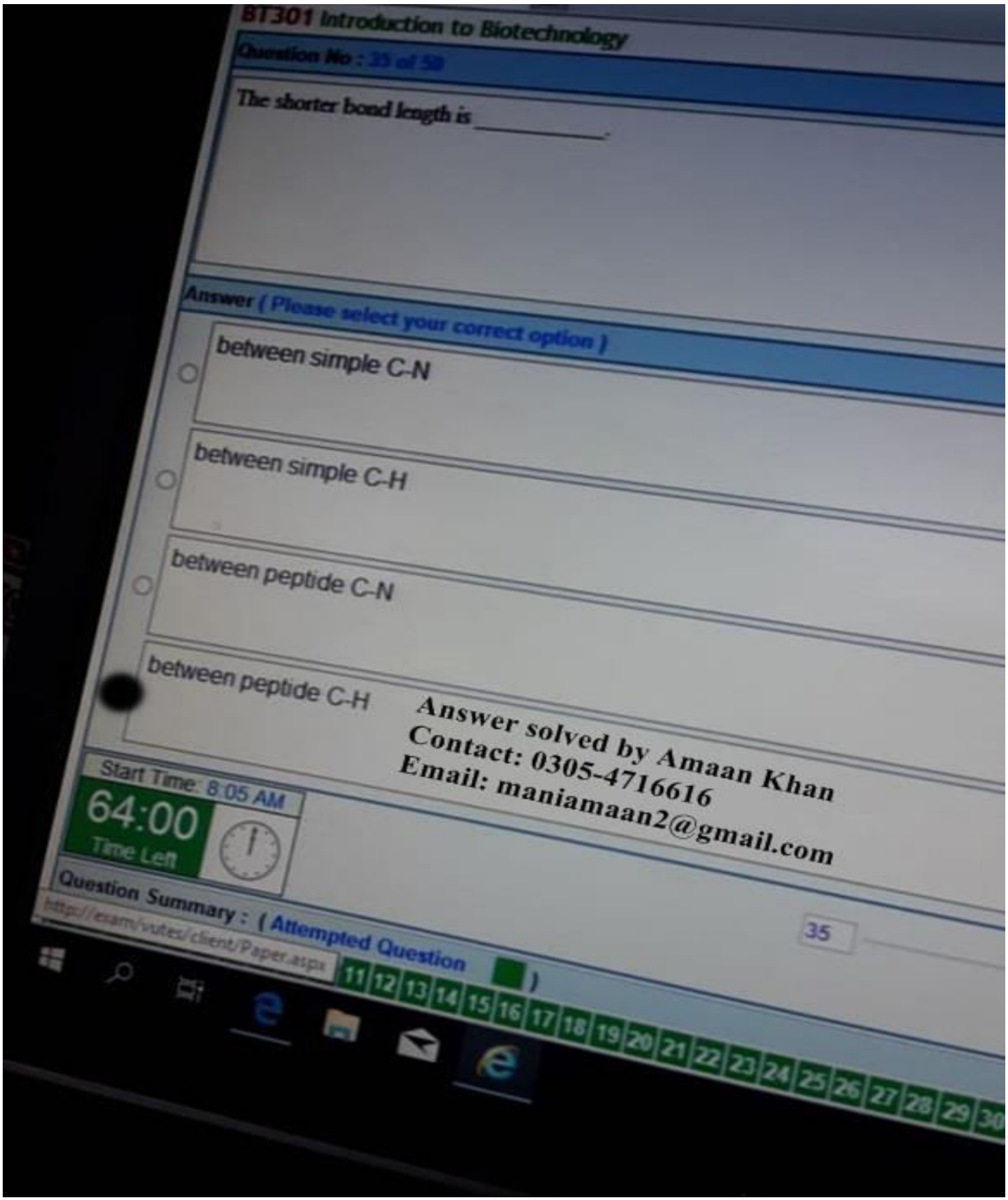
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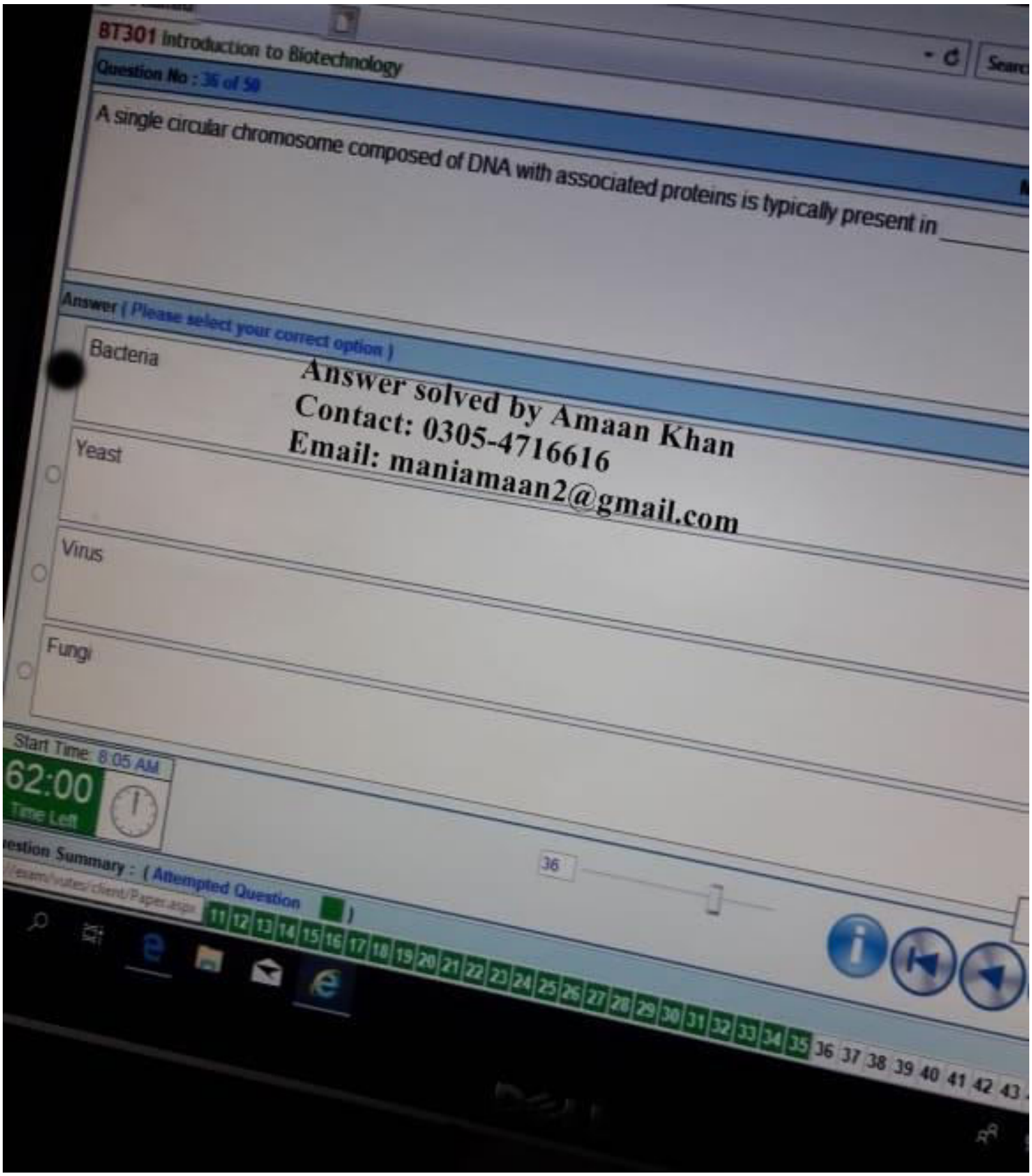
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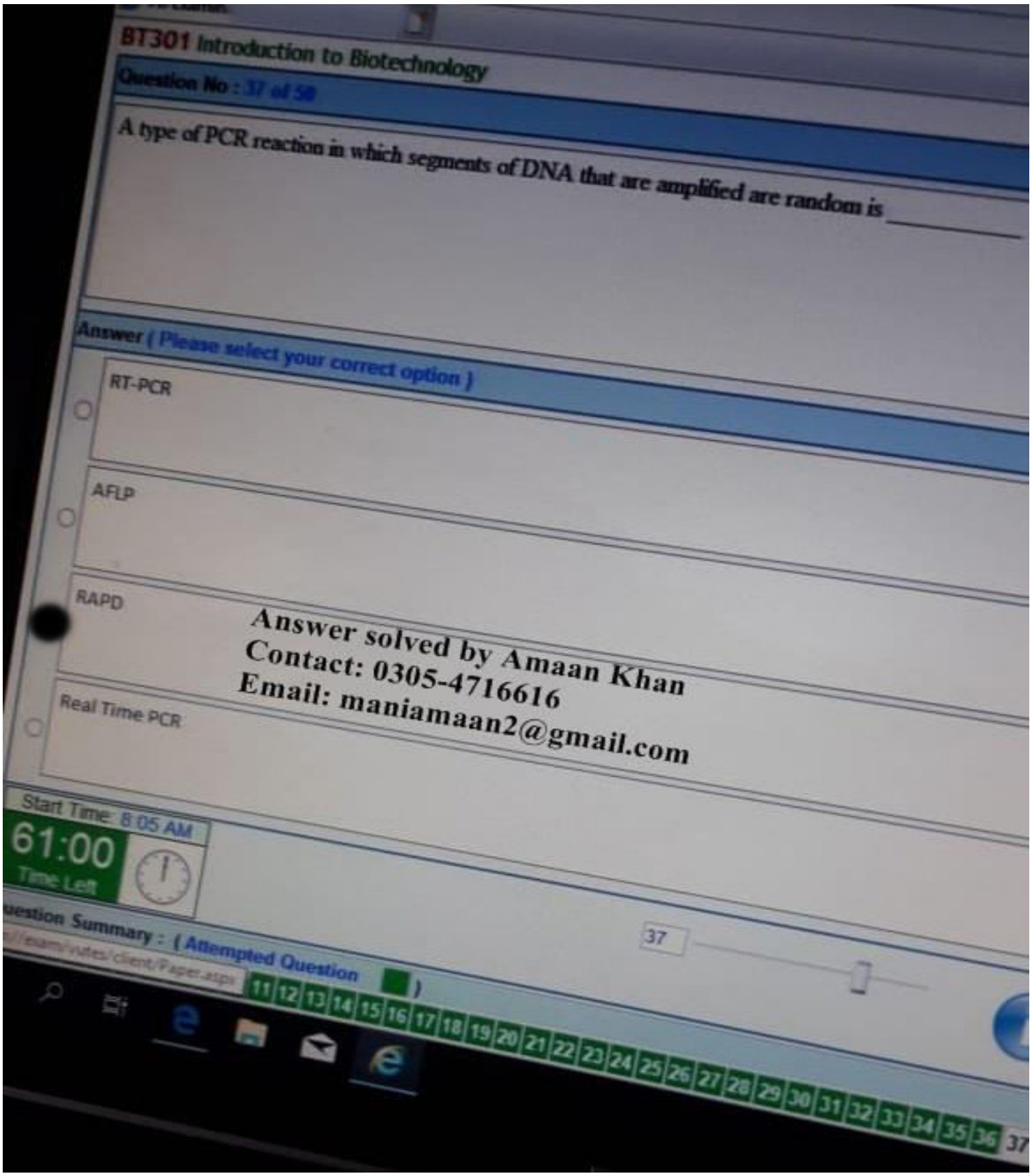
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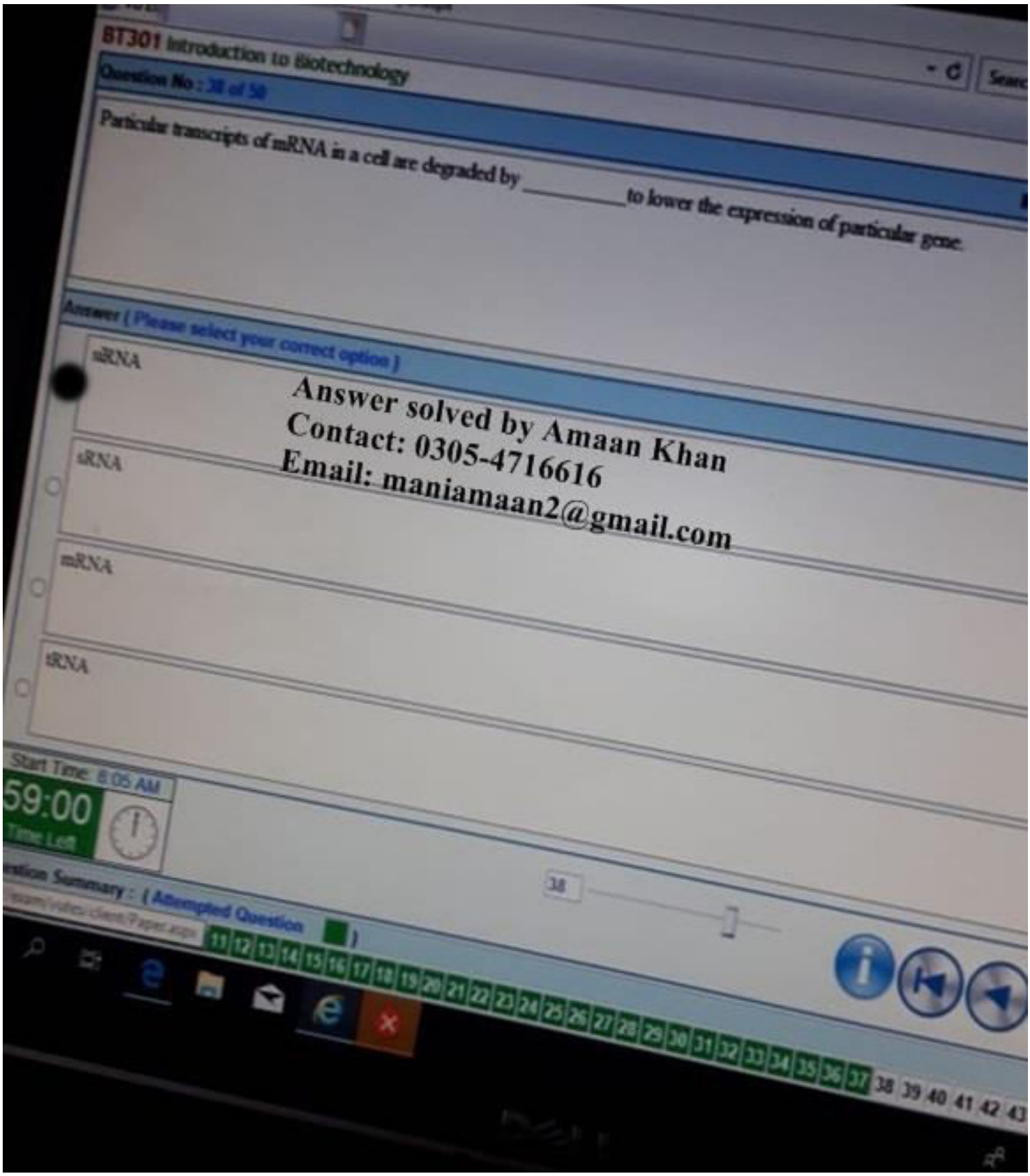
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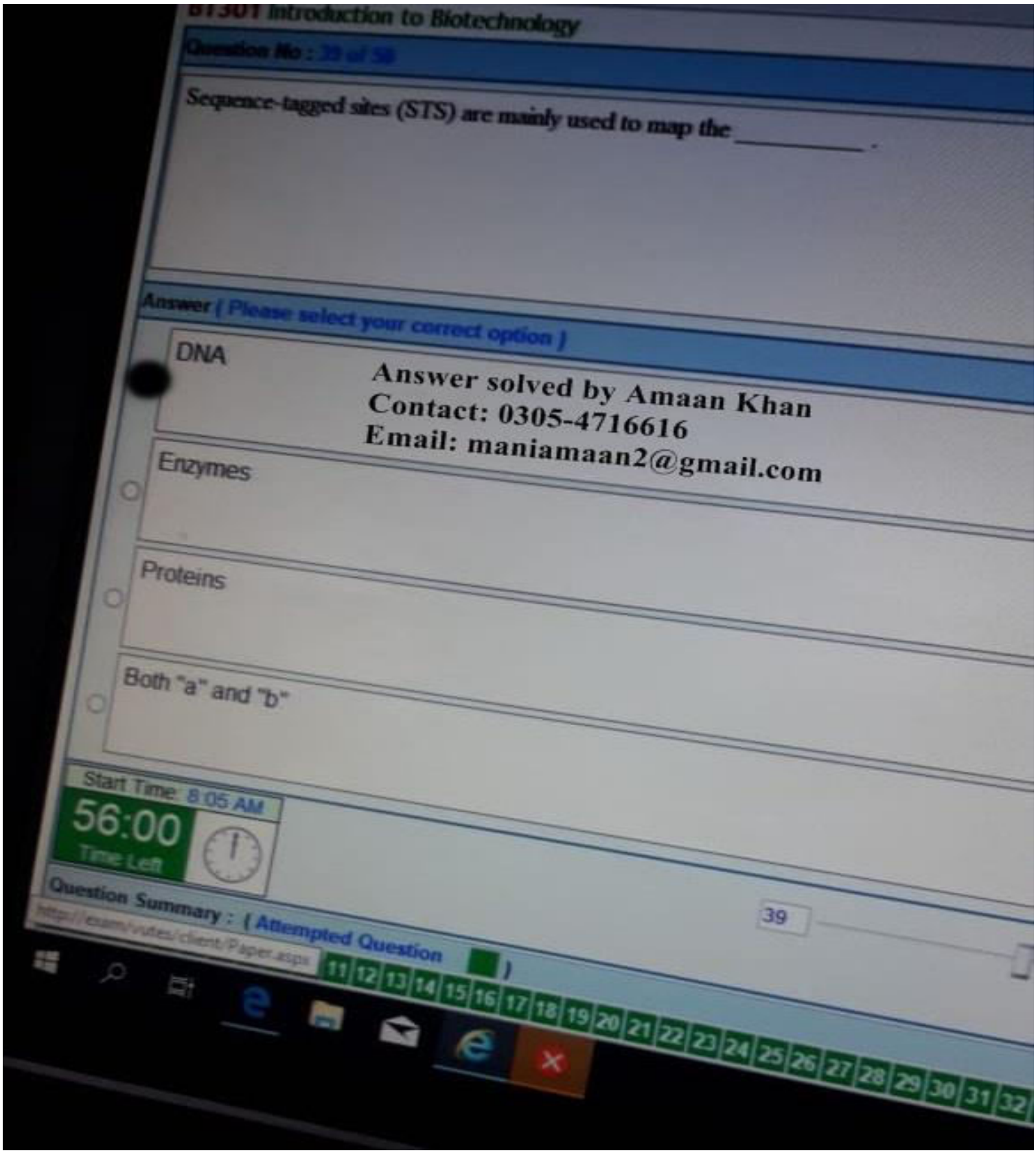
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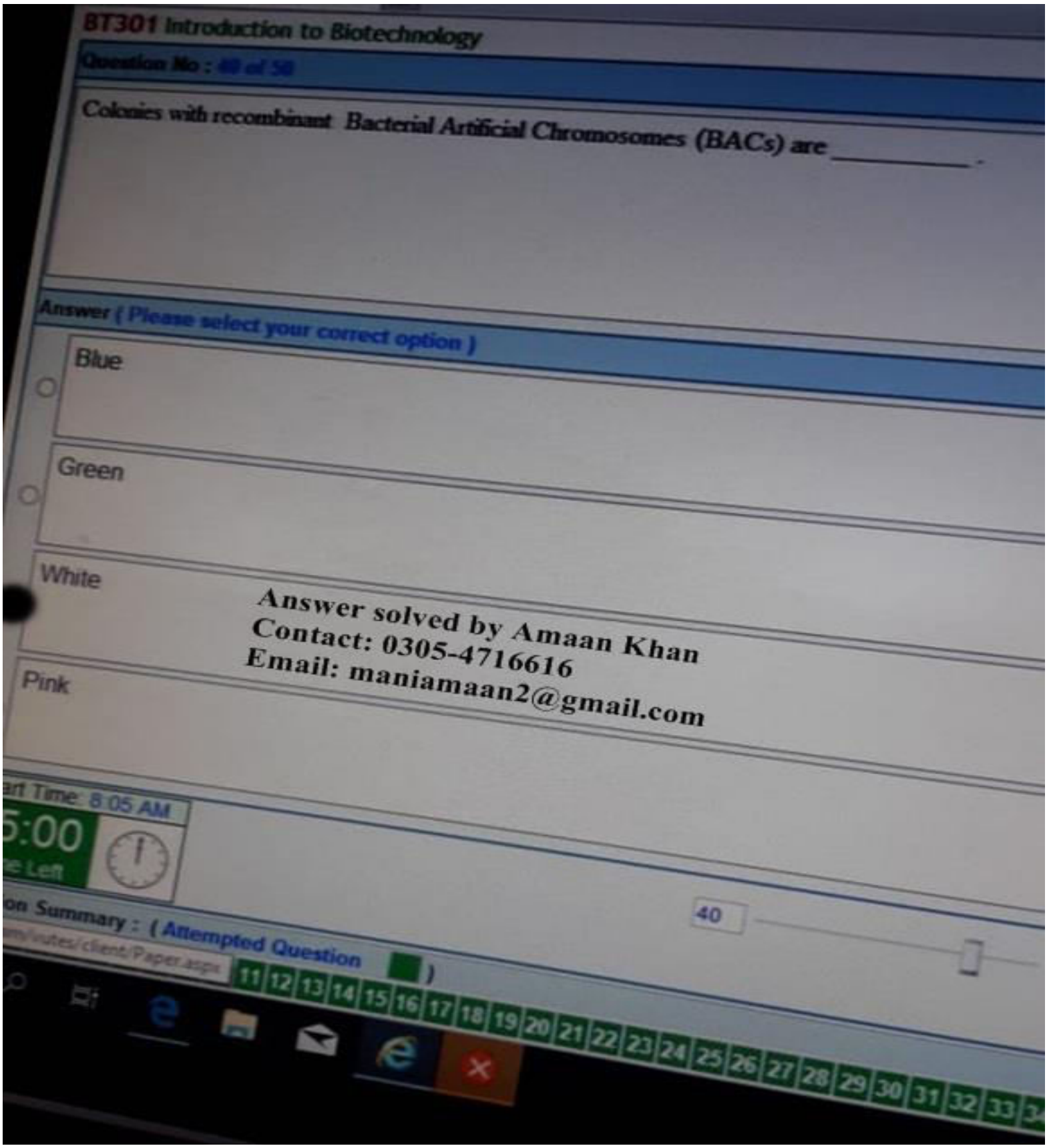
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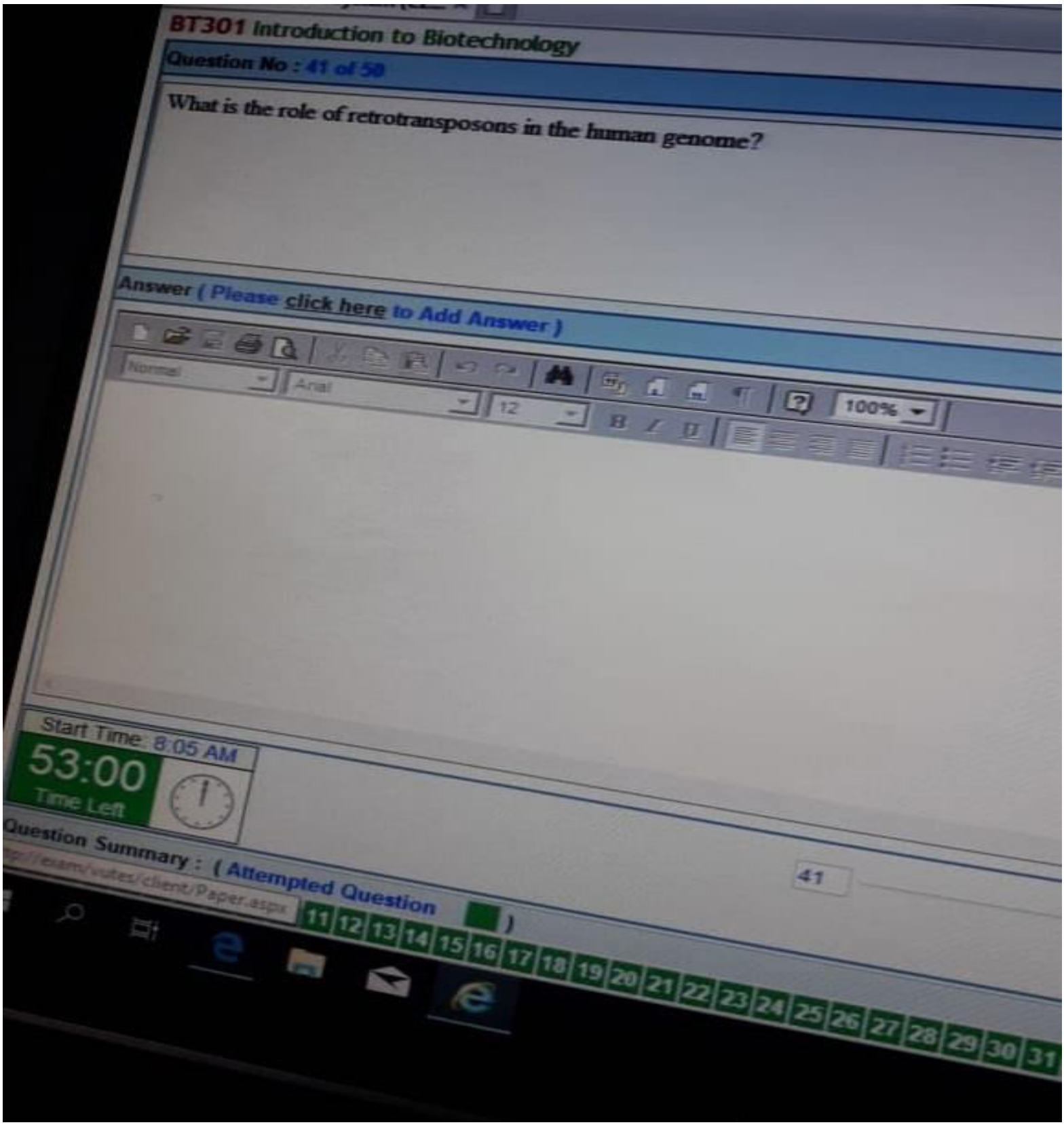
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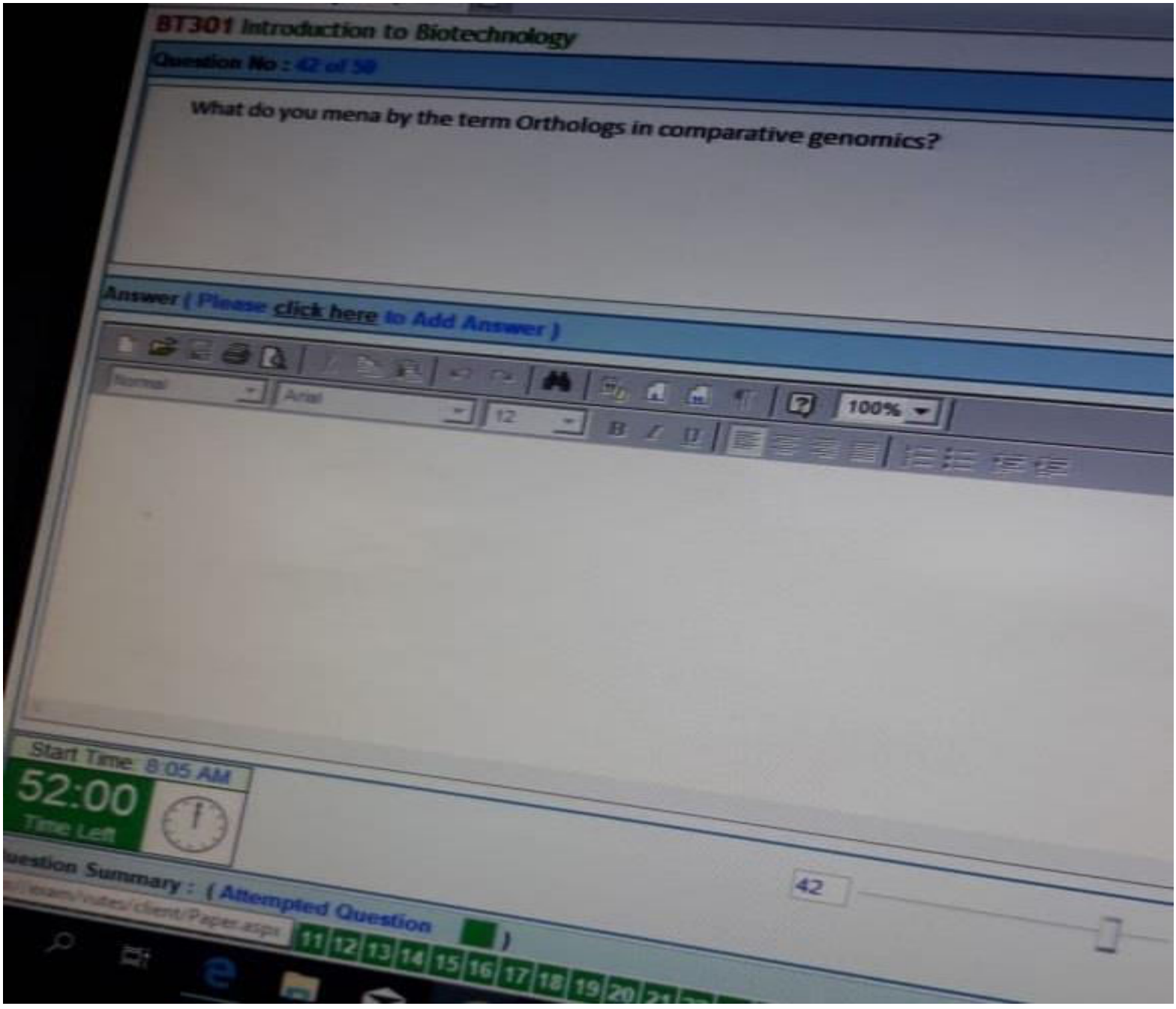
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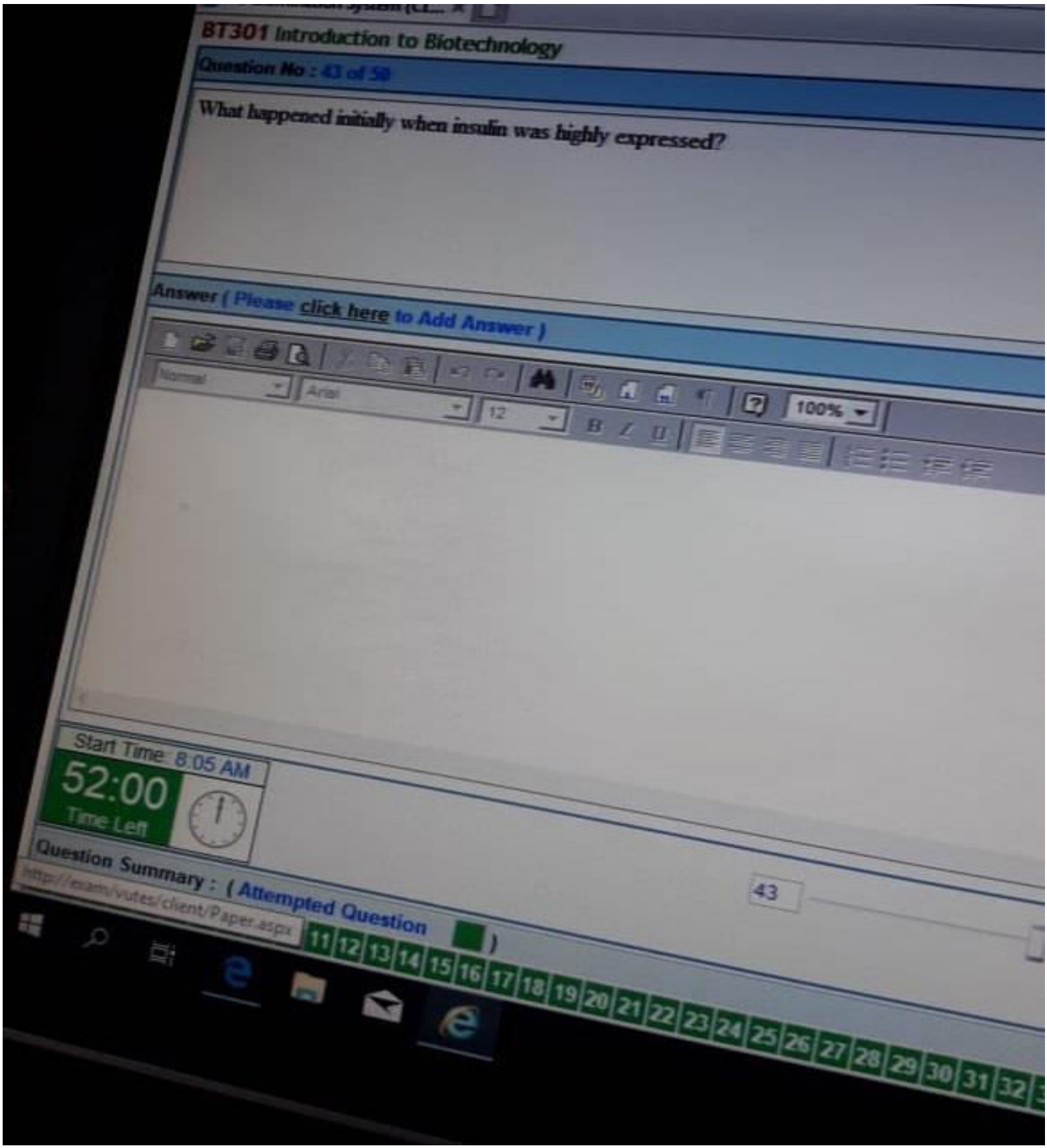
Retrotransposons constitute over 40% of the human genome and consist of several millions of family members. They play important roles in shaping the structure/evolution of the genome.



Orthologs are homologous genes in different organisms that encode proteins with the same function and which have evolved by direct vertical descent.

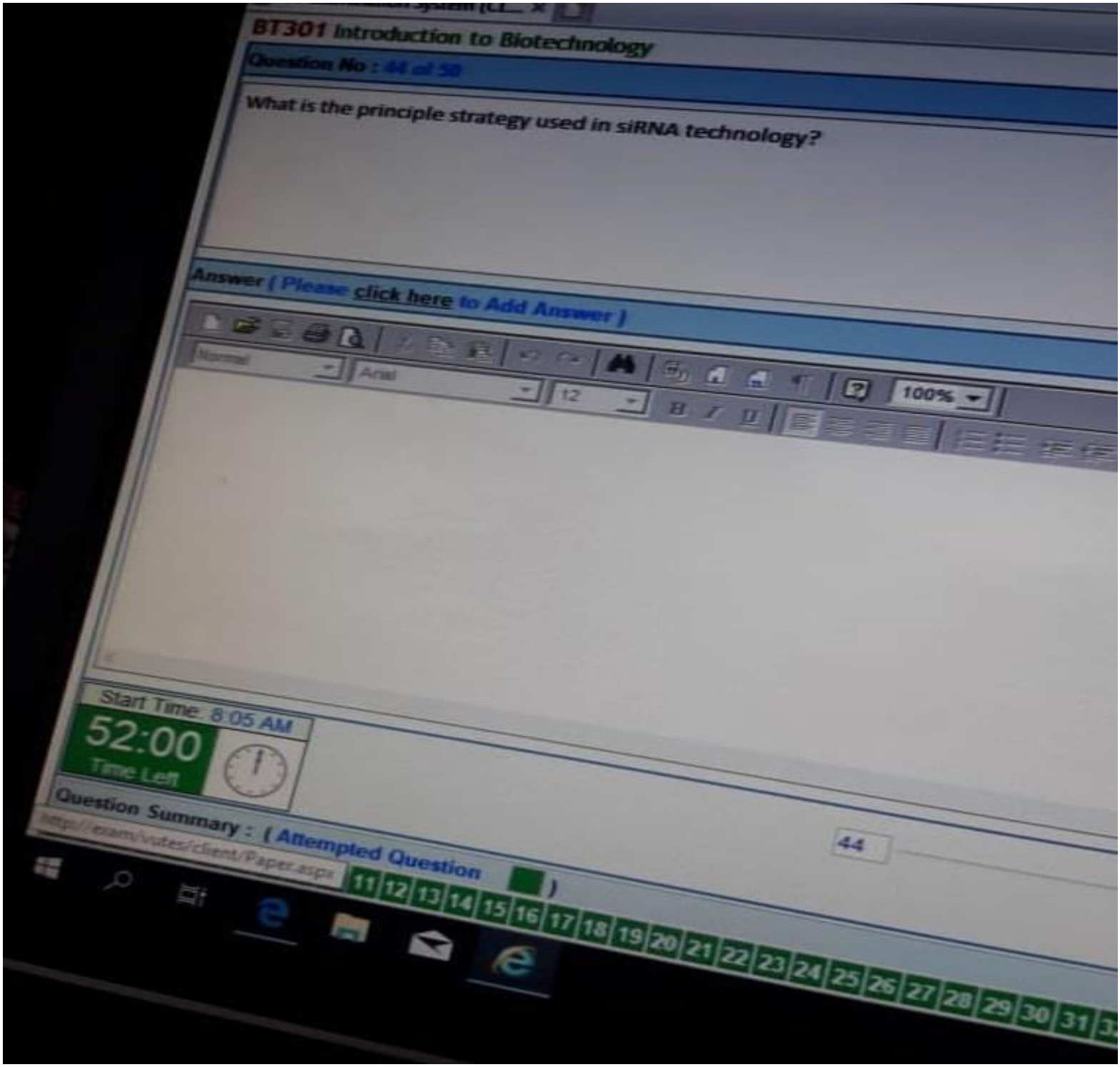
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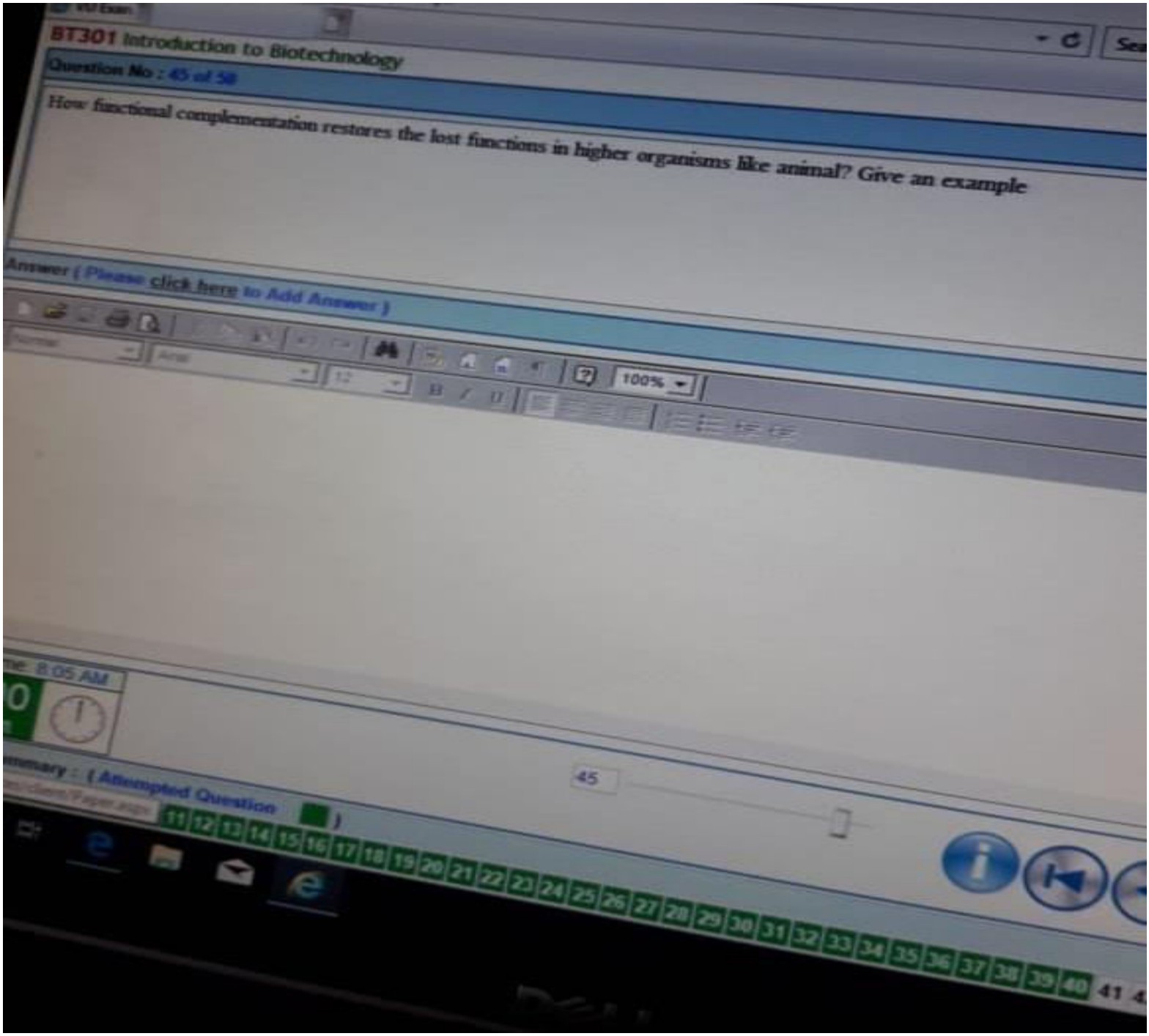
VU MEDICAL ZONE



Small interfering RNA (siRNA) is the most commonly used RNA interference (RNAi) tool for inducing shortterm silencing of protein coding genes. siRNA is a synthetic RNA duplex designed to specifically target a particular mRNA for degradation.

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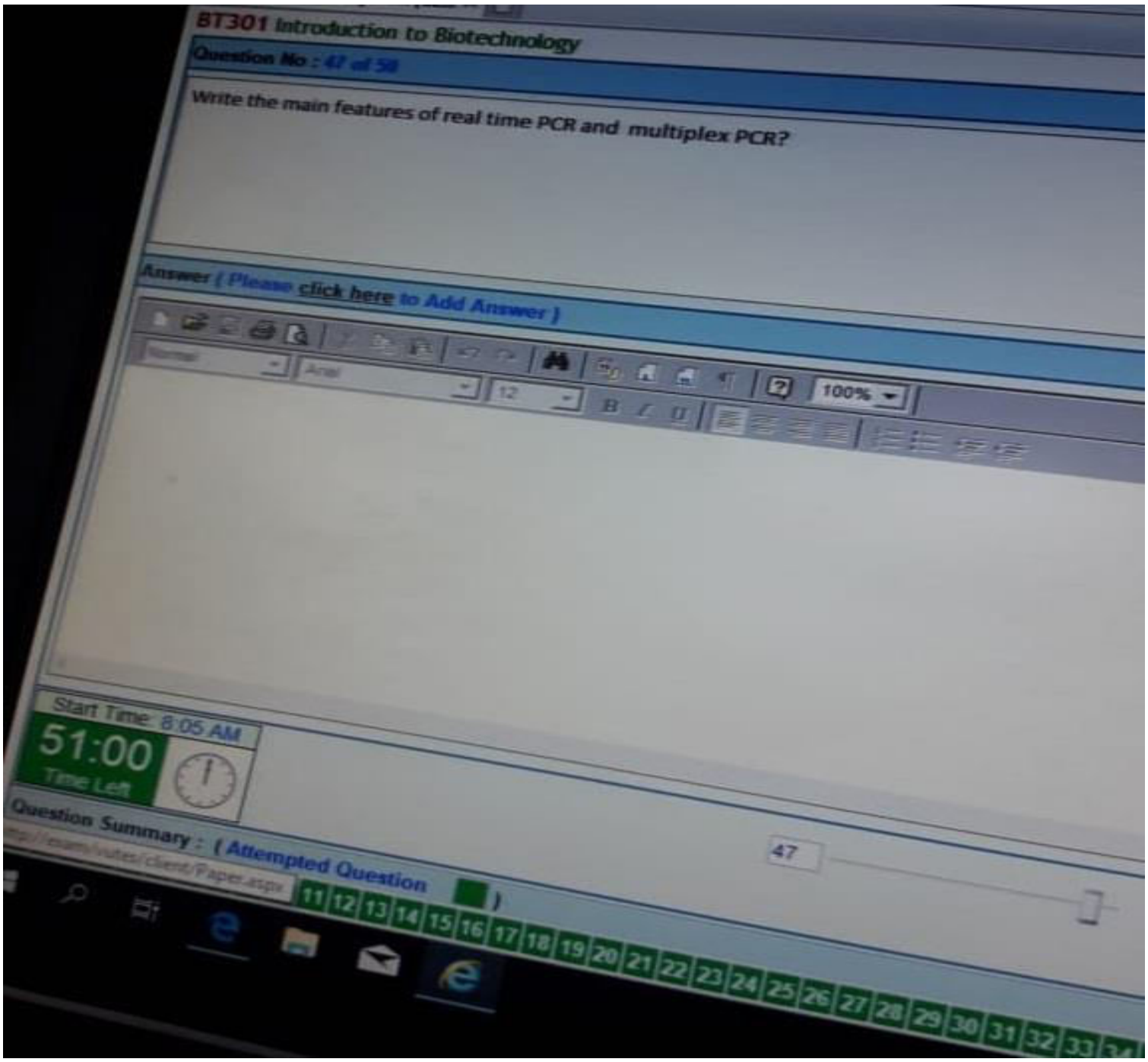
VU MEDICAL ZONE



Functional complementation is the process by which a particular DNA sequence compensates for a missing function in a mutant cell and thus restores the wild type phenotype. Functional complementation is also possible in transgenic animals and plants. It has been used for complementation in transgenic mice to isolate the Shaker-2 gene.

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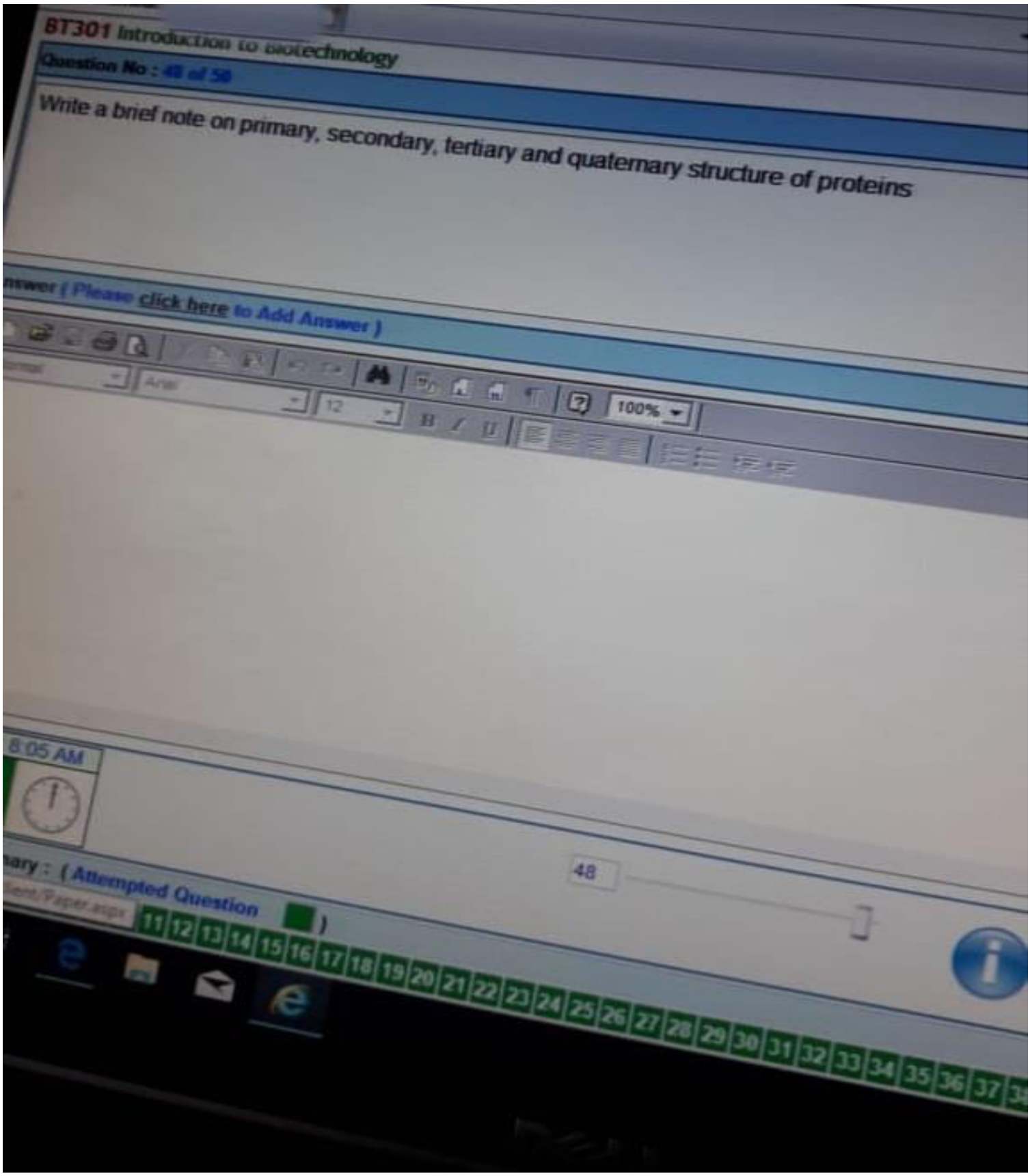


Multiplex PCR Multiplex PCR is used to amplify several different DNA sequences simultaneously.

Real-time quantitative PCR (qPCR) qPCR is used to measure the quantity of a target sequence. It is used to measure starting amounts of DNA, cDNA, or RNA.

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Primary structure

Primary structure or covalent structure of protein refers to the amino acid sequence of its polypeptide chain. Each type of protein has a unique amino acid sequence. Proteins are different from one another on the basis of their amino acid sequence. Peptide Bond Is Rigid and Planar Linus Pauling and Robert Corey carefully analyzed the peptide bond. Their findings laid the foundation for our present understanding of protein structure. They demonstrated that the peptide C - N bond is somewhat shorter than the C - N bond in a simple amine.

The six atoms of the peptide group are co-planar i.e., lie in a single plane, with the oxygen atom of the carbonyl group and the hydrogen atom of the amide nitrogen trans to each other.

Secondary structure

Secondary structure of proteins refers to the local conformation of some part of a polypeptide. The most A few types of secondary structures are particularly stable and occur widely in proteins. Prominent are:-

1. -Helix A polypeptide chain is helically coiled around an imaginary axis and it makes -helix. It is a prominent structure in the proteins.
2. β- conformations Polypeptide chains stretch side by side longitudinally and form sheets. These sheets are called β – sheets.

Tertiary structure

The overall three-dimensional arrangement of all atoms in a protein is referred to as the protein's tertiary structure.

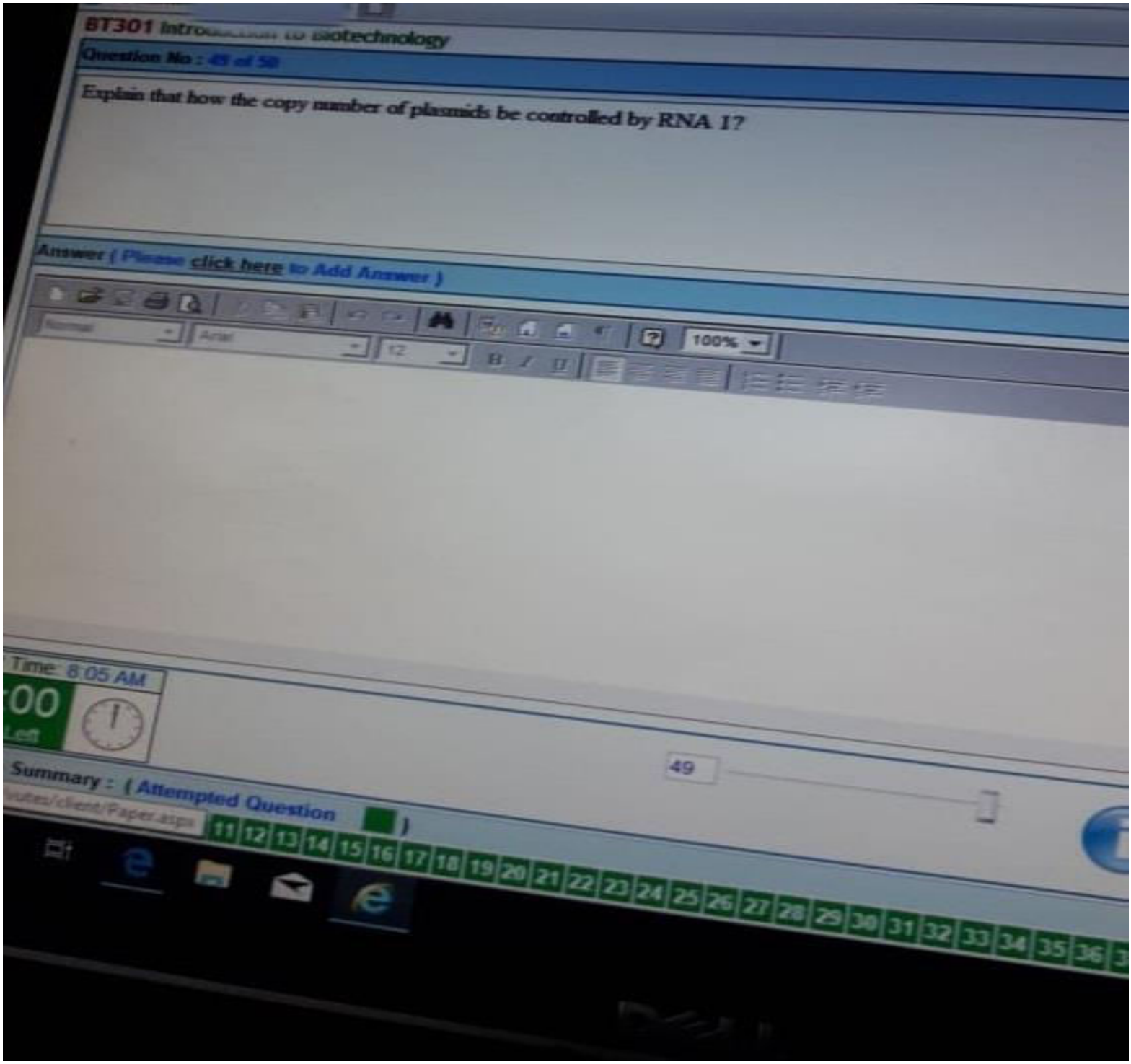
It includes longer-range aspects of amino acid sequence. Amino acids that are far apart in the polypeptide chain may interact within the completely folded structure of a protein. Interacting segments of polypeptide chains are held in their characteristic tertiary positions by different kinds of weak interactions (and sometimes by covalent bonds) between the segments. Large polypeptide chains usually fold into two or more globular clusters known as domains, which often give these proteins a bi- or multilobal appearance.

Quaternary Structure

Some proteins contain two or more separate polypeptide chains or subunits, which may be identical or different. The spatial arrangement of these subunits is known as a protein's quaternary structure. A multi-subunit protein is also referred to as a multimer. A multimer with just a few subunits is called as oligomer and a single subunit or a group of subunits, is called a protomer

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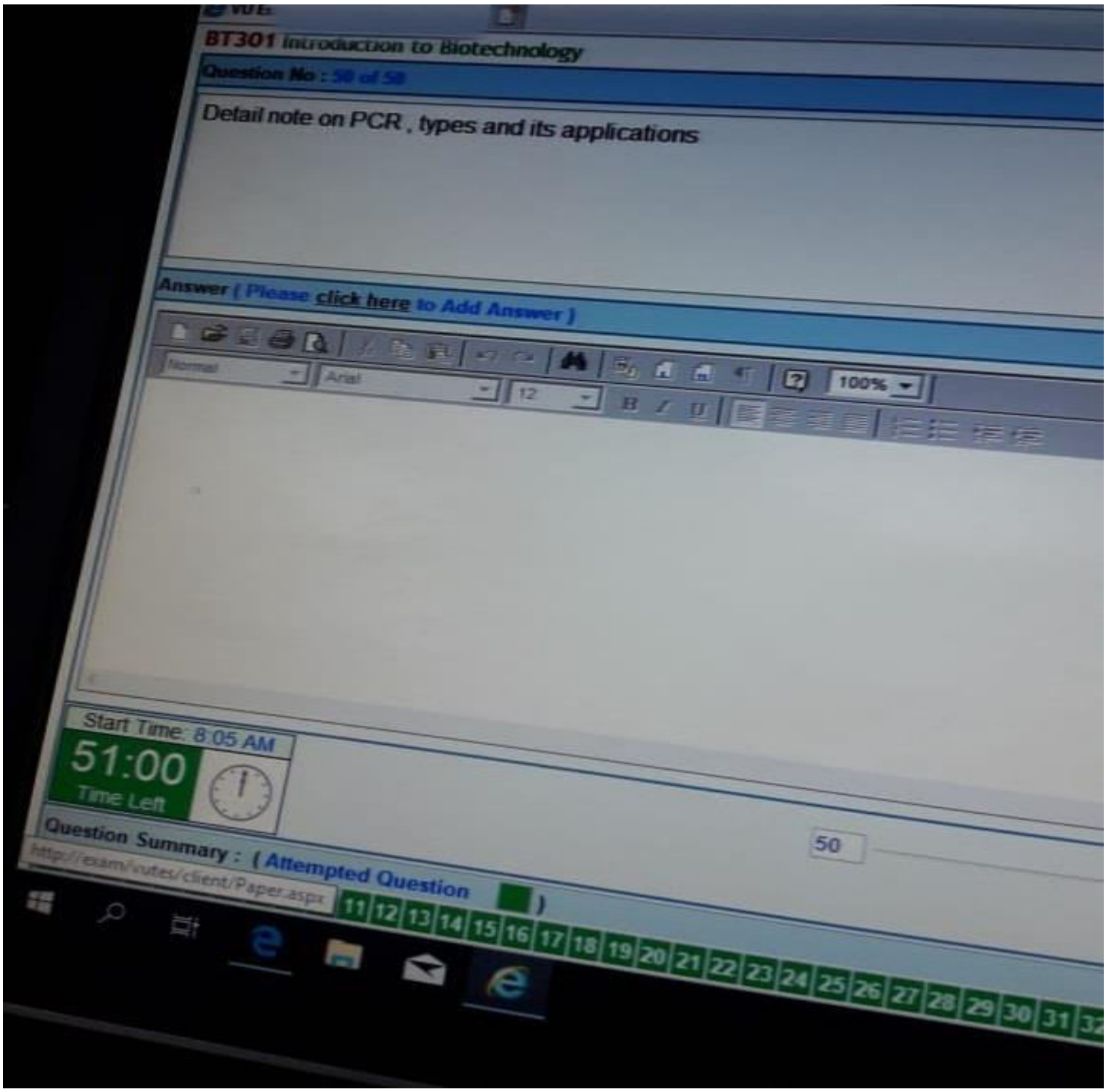


The effect of plasmid copy number:

The number of ribosomes in a cell far exceeds any one class of mRNA. One way of increasing the expression of a cloned gene is to increase the number of the corresponding transcript.

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aquaticus and is used in PCR to amplify DNA fragments

Lecture 158

Basic PCR reaction:

- PCR is a technology in molecular biology used to amplify a single or a few copies of a target DNA across several orders of magnitude, generating millions of copies of a particular DNA sequence.

- PCR can also be used to generate DNA fragments for gene cloning by using primers with specific restriction sites

Table. Comparison of DNA polymerases

of E. coli

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Types of PCR

- Recently, variants have been developed from basic PCR method to improve performance and specificity and to amplify other molecules such as RNA

RT-PCR

- Reverse transcriptase PCR (RT-PCR) was designed to amplify RNA sequences (mainly mRNA) through synthesis of cDNA by reverse transcriptase

- Various strategies can be adopted for first-strand cDNA synthesis

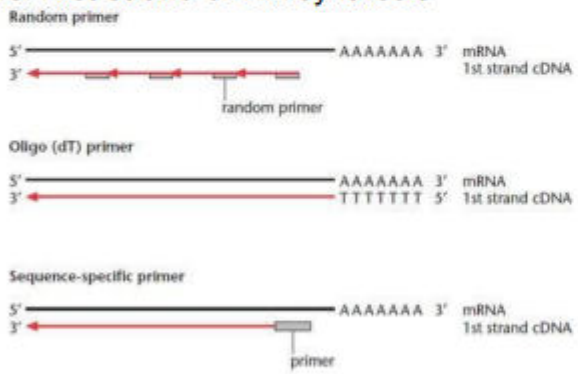


Fig. Strategies for synthesis of first-strand cDNA

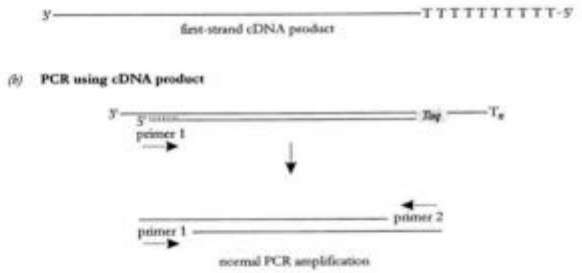
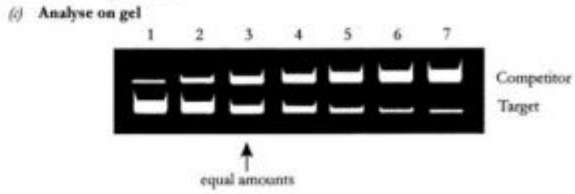


Fig. RT-PCR

Lecture 164

Competitor RT-PCR

- One use of RT-PCR is in determining the amount of mRNA in a sample (competitor RT-PCR) **RT-PCR**
- Reverse transcriptase PCR (RT-PCR) was designed to amplify RNA sequences (mainly mRNA) through synthesis of cDNA by reverse transcriptase



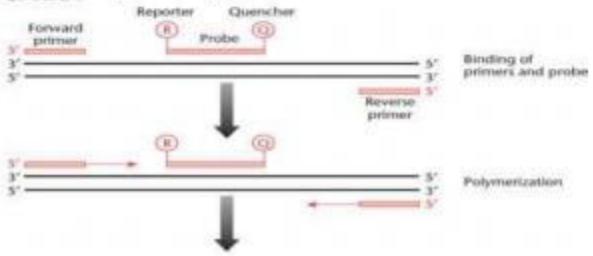
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Lecture 165

Real-time quantitative PCR (qPCR)

- qPCR is used to measure the quantity of a target sequence
- It is used to measure starting amounts of DNA, cDNA, or RNA

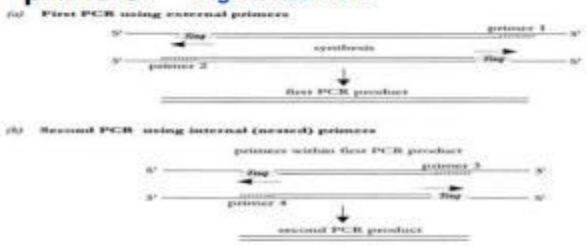


Lecture 166

Nested PCR:

- In nested PCR, two sets of primers are used in two successive PCRs
- First set of primers allows a **first amplification**
- Product of first PCR is subjected to a second PCR using the **second set of primers**.

Fig. Nested PCR



Lecture 167

Inverse PCR:

- Inverse PCR is used to identify the flanking sequences around genomic inserts.

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Lecture 168

Multiplex PCR

- Multiplex PCR is used to amplify several different DNA sequences simultaneously.

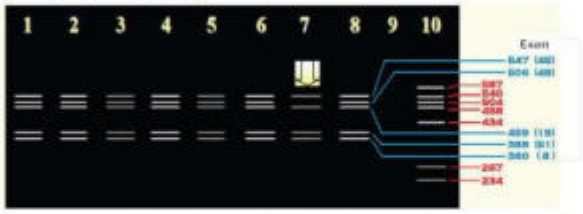


Fig. Results of multiplex PCR in a patient with Duchenne Muscular Dystrophy Inverse PCR Hernandex-Rodriguez et al., 2000; Hernandex-Rodriguez & Restrepo, 2002 (www.intechopen.com)

Lecture 169

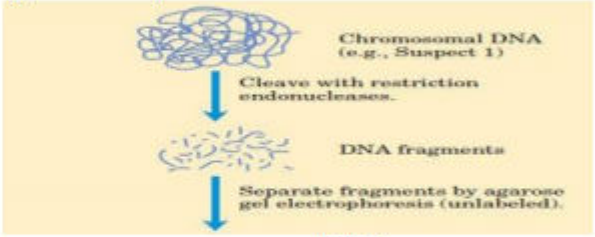
RAPD:

- RAPD-Random amplification of polymorphic DNA is a type of PCR

Lecture 170

RFLP:

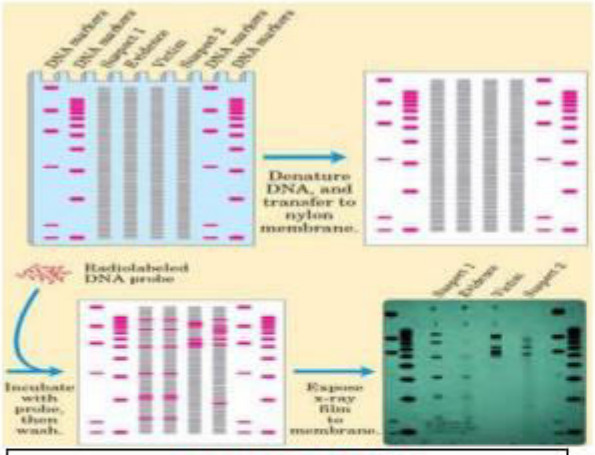
- RFLP-Restriction fragment length polymorphism exploits variations in homologous DNA sequences
- In RFLP, DNA sample is digested by restriction enzymes and separated by gel electrophoresis



DNA fingerprinting

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Lecture 171

AFLP-PCR:

- AFLPs-Amplified fragment length polymorphisms are differences in restriction fragments length caused by SNPs or INDELS
- AFLP is a PCR based tool used in genetic engineering.

Lecture 172

Applications of PCR

PCR has widespread applications in various fields of life sciences including genetic engineering, medical, forensic, agriculture, environment etc. **PCR-Gene cloning and expression** PCR has been used in gene cloning and screening of genomic libraries

Lecture 173

PCR Medicine: PCR has major impact

Molecular tools have also allowed to perform prenatal genetic diagnosis.

Lecture 174

PCR-Forensic sciences: Forensic science is the application of scientific procedures to solve criminal & legal matters

Molecular methods are used to established the filiations of a person or to obtain evidence from minimal samples of saliva, semen or other tissues. **PCR-DNA profiling:** DNA profiling or DNA fingerprinting is a forensic technique used to identify individuals by characteristics of their DNA

Lecture 175

PCR-Agricultural sciences and environment PCR has also facilitated research in detection of pathogens in plants, animals and environment.

Lecture 176

PCR-Molecular paleontology

Molecular paleontology refers to the recovery and analysis of DNA and protein from ancient human, animal and plant remains.

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